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PHYSICS

1) The internal and external diameter of a hollow cylinder are measured with the help of a Vernier callipers. Their values are 4.23 ± 0.01 cm and 3.87 ± 0.01cm, respectively. The thickness of the wall of the cylinder is

(1) 0.36 ± 0.02 cm

(2) 0.18 ± 0.02 cm

(3) 0.36 ± 0.01 cm

(4) 0.18 ± 0.01 cm

2) A wire has a mass $(0.3 \pm 0.003)\ g$, radius $(0.5 \pm 0.005)\ mm$ and length $(6 \pm 0.06)\ cm$. Then maximum % error in measurement of density is :

(1) 1%

(2) 2%

(3) 3%

(4) 4%

3) Complete the following calculations, reporting your answer to the correct number of significant figures :

$$\frac{10,569 - 958.82 - 1.8 \times 10^{-1}}{0.9950 + 8.60 + 0.015}$$

(1) 1000

(2) 1×10^3

(3) 1.00×10^3

(4) All of these

4) If $F = \frac{2\pi C r e^{-kr}}{v^2}$ where F is force, r is distance, v is velocity and C & k are constants then dimension of C is :-

(1) Dimensionless

(2) MLT^{-2}

(3) ML^2T^{-3}

(4) ML^2T^{-4}

5) The speed v of an object is given by equation $v = At^3 - Bt$, where t refers to time :-

(a) The dimensions of A is LT^{-4}.

(b) The dimensions of B is LT^{-2}.

(c) The time at which object is in equilibrium (acceleration = 0) is given by $t = \sqrt{\frac{B}{3A}}$.

(d) The time at which object is in equilibrium (acceleration = 0) is given by $t = \sqrt{\frac{A}{3B}}$.

(1) a, b

(2) a, b, c

(3) a, b, d

(4) a, b, c, d

6) Which of the following quantities cannot enter into the list of fundamental quantities in any system of units ?

(1) Mass, time and acceleration

(2) Length, volume and linear momentum

(3) Mass, velocity and time

(4) Length, force and time

7) $\vec{A} + \vec{B}$ is perpendicular to $\vec{A}$ and it is given that $\left|\vec{B}\right| = 2\left|\vec{A} + \vec{B}\right|$ then angle between $\vec{A}$ and $\vec{B}$ will be :-

(1) 30°

(2) 150°

(3) 120°

(4) 0°

8) What is the magnitude of projection of $\vec{P} = -5\hat{i} + 8\hat{j} + 6\hat{k}$ on yz plane ?

(1) $5\sqrt{5}$

(2) 10

(3) 6

(4) 8

9) The least count of the main scale of a screw gauge is 1mm. The minimum number of divisions on its circular scale required to measure 8 μm diameter of a wire is :

(1) 50

(2) 200

(3) 100

(4) 125

10) A heavy box is to dragged along a rough horizontal floor. To do so, person A pushes it at an angle 30° from the horizontal and requires a minimum force F_A, while person B pulls the box at an angle 60° from the horizontal and needs minimum force F_B. If the coefficient of friction between the box and the floor is $\frac{\sqrt{3}}{5}$, the ratio $\frac{F_A}{F_B}$ is :

(1) $\sqrt{3}$

(2) $\frac{5}{\sqrt{3}}$

(3) $\sqrt{\frac{3}{2}}$

(4) $\frac{2}{\sqrt{3}}$

11) A child weighing 25 kg slides down a rope hanging from the branch of a tall tree. If the force of friction acting against him is 2N, ... m/s^2 is the acceleration of the child (Take g = 9.8 m/s^2) :

(1) 22.5

(2) 8

(3) 5

(4) 9.72

12) A force of 750 N is applied to a block of mass 102 kg to prevent it from sliding on a plane with an inclination angle 30° with the horizontal. If the coefficients of static friction and kinetic friction between the block and the plane are 0.4 and 0.3 respectively, then the frictional force acting on the block is ..... N : (g = 9.8 m/s^2)

(1) 750

(2) 500

(3) 345

(4) 250

13) A spaceship in space sweeps stationary interplanetary dust. As a result, its mass increases at a rate $\frac{dM(t)}{dt} = bv^2(t)$, where v(t) is its instantaneous velocity. The instantaneous acceleration of the satellite is :-

(1) $-\frac{2bv^3}{M(t)}$

(2) $-\frac{bv^3}{2M(t)}$

(3) $-bv^3(t)$

(4) $-\frac{bv^3}{M(t)}$

14) A player kicks a football of mass 0.5 kg and the football begins to move with a velocity of 10 m/s. If the contact between the leg and the football lasts for $\frac{1}{50}$ sec, then the force acted on the football should be ..... N : (g = 9.8 m/s^2)

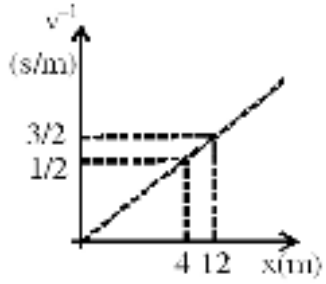
(1) 2500

(2) 1250

(3) 250

(4) 625

15) Graph of 1/v versus x for a particle under motion is shown as, where v is velocity and x is position. The time taken by particle to move from x = 4 to x = 12 m is:



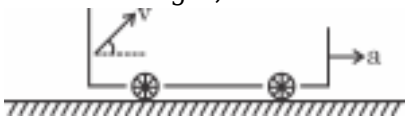
(1) 16/3 sec.

(2) 10 sec.

(3) 8 sec.

(4) 12 sec.

16) A particle is projected from trolley car with a velocity $\vec{v}$. If trolley car moves with an acceleration $\vec{a}$ towards right, which of the following remain unchanged relative to both ground and trolley car ?



(1) Range

(2) Maximum range

(3) Horizontal velocity

(4) Time of flight

17) The velocity of a moving particle is given by $\vec{v} = \sin\left(\frac{3\pi}{10}t\right)\hat{i} + \cos\left(\frac{3\pi}{10}t\right)\hat{j}$. The distance travelled by the particle in first second is

(1) $\sin^{-1}\left(\cos\frac{3\pi}{10}\right)$ units

(2) $\cos^{-1}\left(\sin\frac{3\pi}{10}\right)$ units

(3) 2 units

(4) 1 units

18) A projectile is projected with velocity of 25 m/s at an angle θ with the horizontal. After t seconds its inclination with horizontal becomes zero. If R represents horizontal range of the projectile, the value of θ will be : [use g = 10 m/s^2]

(1) $\frac{1}{2}\sin^{-1}\left(\frac{5t^2}{4R}\right)$

(2) $\frac{1}{2}\sin^{-1}\left(\frac{4R}{5t^2}\right)$

(3) $\tan^{-1}\left(\frac{4t^2}{5R}\right)$

(4) $\cot^{-1}\left(\frac{R}{20t^2}\right)$

19)

A man with some passengers in his boat, starts perpendicular to flow of river 200 m wide and flowing with 2 m/s. Speed of boat in still water is 4 m/s. When he reaches half the width of river the passengers asked him that they want to reach the just opposite end from where they have started. How many times more time (total), it would take to that if he would have denied the passengers ?

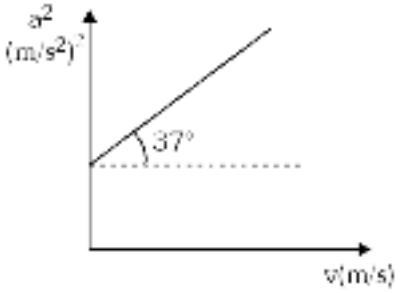
(1) $\frac{4}{3}$

(2) $\frac{3}{4}$

(3) $\frac{3}{5}$

(4) $\frac{5}{3}$

20) In the graph shown between a^2 vs v where a is the acceleration v is the velocity of particle. If rate of change of acceleration is $\frac{3}{x}$ when velocity of particle will be 5 m/s. Find x.



(1) 5

(2) 3

(3) 8

(4) 16

21) Two particles are moving with velocity v_1 and v_2 respectively, which of the following is correct about relative velocity? (v_1, $v_2 \neq 0$)

(1) Velocity of 1 w.r.t. 2 can not be equal to $|v_1|$.

(2) Velocity of 1 w.r.t. 2 can be smaller than $|v_1| - |v_2|$.

(3) Velocity of 1 w.r.t. 2 can not be greater than $|v_1| + |v_2|$.

(4) Velocity of 1 w.r.t. 2 can not be equal to $|v_2|$.

22) A horizontal wind is blowing with a constant velocity v towards north east. A man start from rest with constant acceleration 'a' towards north. The time after which man will feel the wind blowing towards east is t. If $\frac{v}{a} = 2\sqrt{2}$. Find the value of t.

(1) $\sqrt{2}$

(2) 2

(3) $2\sqrt{2}$

(4) $\frac{1}{2}$

23) A ball is projected upward from the foot of a tower. The ball crosses the top of the tower twice after an interval of t seconds and ball reaches the ground after 2t seconds If height of the tower is 135 m. Find the value of t in seconds. (Take g = 10m/s^2)

(1) 3 sec

(2) 4 sec

(3) 6 sec

(4) 9 sec

24) Position vector of a particle is given by $\vec{r} = at\hat{i} + bt^2\hat{j}$. Its average velocity is first t seconds will be :

(1) $a\hat{i} + \frac{bt}{2}\hat{j}$

(2) $a\hat{i} + bt\hat{j}$

(3) $\frac{a\hat{i}}{2} + \frac{bt}{3}\hat{j}$

(4) $a\hat{i} + 2bt\hat{j}$

25) Instantaneous velocity of an object varies with time as v = α – βt^2. Find maximum positive displacement from origin assuming α, β are positive constants.

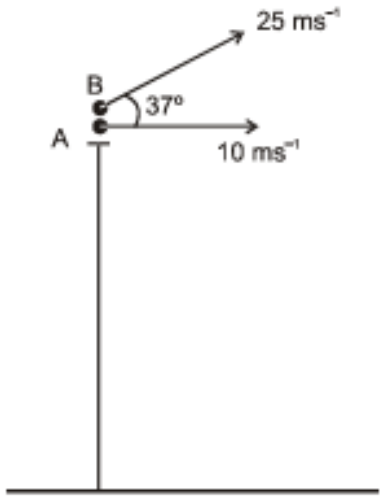
(1) $\frac{\alpha}{3}\sqrt{\frac{\alpha}{\beta}}$

(2) $2\alpha\sqrt{\frac{\alpha}{\beta}}$

(3) $\frac{2}{3}\alpha\sqrt{\frac{\alpha}{\beta}}$

(4) $\sqrt{\frac{\alpha\beta}{3}}$

26) Particle A and B are projected at t = 0 from a high tower then which of following statement is wrong about their motion when both particles are in air :



(1) their relative velocity is non zero constant

(2) separation between two particle remains constant

(3) distance between two particle at t = 2 sec is $10\sqrt{13}$ m

(4) acceleration of particle with respect to each other is zero.

27) The displacement 'x' and time of travel 't' for a particle moving on a straight line are related as t = $\sqrt{(x+1)(x-1)}$. Its acceleration at time t is

(1) $\frac{1}{x}-\frac{1}{x^2}$

(2) $\frac{1}{x^3}$

(3) $\frac{-t^2}{x^3}$

(4) $\frac{-t}{x^2}$

28) Two projectiles are projected at angles (θ) and $\left(\frac{\pi}{2}-\theta\right)$ to the horizontal respectively with same speed 20 m/s. One of them rises 10 m higher than the other. Find the angles of projection. (Take g = 10 m/s^2)

(1) 37°, 53°

(2) 30°, 60°

(3) 25°, 55°

(4) 18.5°, 71.5°

29) The diameter of a cylinder is measured using a Vernier callipers with no zero error. It is found that the zero of the Vernier scale lies between 5.10 cm and 5.15 cm of the main scale. The Vernier scale has 50 divisions equivalent to 2.45 cm. The 24th division of the Vernier scale excatly coincides with one of the main scale divisions. The diameter of the cylinder is :-

(1) 5.112 cm

(2) 5.124 cm

(3) 5.136 cm

(4) 5.148 cm

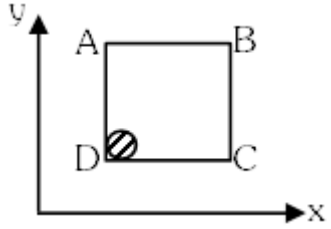
30) The density of a material in CGS system of units is 4 g/cm^3. In a system of units in which unit of length is 10 cm and unit of mass is 100 g, the value of density of material will be :-

(1) 0.04
(2) 0.4
(3) 40
(4) 400

31) The pitch of a screw gauge is 1 mm and there are 50 divisions on its cap. When nothing is put in between the studs. 44th division of the circular scale coincides with the reference line and zero of the main scale is not visible. When a glass plate is placed between the studs, the main scale reads three divisions and the circular scale reads 26 divisions. Calculate the thickness of the plate.

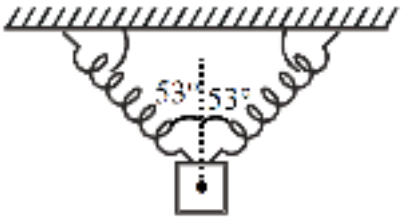
(1) 3.64
(2) 4.64
(3) 3.26
(4) 4.26

32) A solid sphere of mass 2kg is resting inside a cube as shown in figure. The cube is moving with velocity $\vec{v} = (5t\hat{i} + 2t\hat{j})$ m/s. Here 't' is time in second. If all surfaces are smooth and sphere is at rest w.r.t. cube, then total force exerted by the sphere on the cube is : (g = 10 m/s^2)

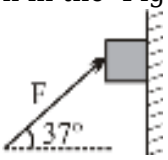


(1) $\sqrt{29}$ N
(2) 29 N
(3) 26 N
(4) $\sqrt{89}$ N

33) The block of mass 2 kg shown is in equilibrium. At t = 0 right spring in figure breaks. Find the instantaneous acceleration of block ?



(1) 5 m/s^2
(2) $\frac{25}{3}$ m/s^2
(3) $\frac{20}{3}$ m/s^2
(4) None

34) A 1kg block is being pushed against a wall by a force F = 75 N as shown in the Figure. The coefficient of friction is 0.25. The magnitude of acceleration of the block is 

(1) 10 m/s^2

(2) 20 m/s^2

(3) 5 m/s^2

(4) none

35) Consider the pulley system in the diagram below. The unknown force F being applied is just sufficient to hold the system in equilibrium. The block has mass M, while the pulleys and ropes have negligible masses. What is the tension T in the upper cable (i.e. the cable connecting the top pulley to the ceiling) in terms of M and the acceleration due to gravity g only. 

(1) $\frac{8Mg}{7}$

(2) Mg

(3) $\frac{Mg}{5}$

(4) $\frac{8Mg}{5}$

36) A 10 kg stone is suspended with a rope of breaking strength 30 kg wt. The minimum time in which the stone can be raised through a height 10 m, starting from rest is (taking g = 10 N/kg) :-

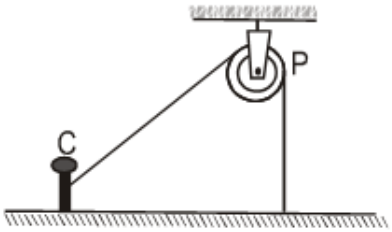
(1) 0.5 seconds

(2) 1.0 seconds

(3) $\sqrt{\frac{2}{3}}$ seconds

(4) 2.0 seconds

37) One end of massless rope, which passes over a massless and frictionless pulley P is tied to a hook C while the other end is free. Maximum tension that the rope can bear is 360 N. With what value of minimum safe acceleration (in ms^{-2}) can a man of 60 kg climb down the rope ?



(1) 16

(2) 6

(3) 4

(4) 8

38) The potential energy of a particle of mass 5 kg moving in the x – y plane is given by U = (–7x + 24y)J, where x and y are given in metre. If the particle starts from rest, from the origin, then the speed of the particle at t = 2 s is

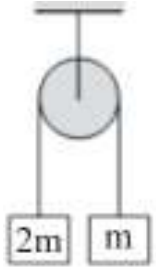
(1) 5 m/s

(2) 14 m/s

(3) 17.5 m/s

(4) 10 m/s

39) The tension in the string is 16N when the system is set into motion. Find the decrease in gravitational potential energy during the first second after the system is released from rest (g = 9.8 m/s^2) :



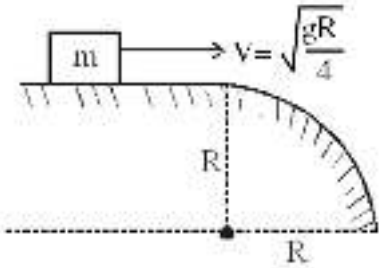
(1) 9.8 J

(2) 19.6 J

(3) 10 J

(4) 20 J

40) A particle slides on frictionless surface then at what angle from vertical it will leave contact from spherical surface. Given that its initial speed is v = $\sqrt{\frac{gR}{4}}$ m/s where R is radius of spherical portion.



(1) $\cos^{-1}\left(\frac{2}{3}\right)$

(2) $\cos^{-1}\left(\frac{3}{2}\right)$

(3) $\cos^{-1}\left(\frac{3}{4}\right)$

(4) $\cos^{-1}\left(\frac{4}{6}\right)$

41) A force given by the relation F = 8t, acts on a body of mass 2 kg, initially at rest. Find the work done by this force on the body during first 2 s of its motion.

(1) 64 J

(2) Zero

(3) –64 J

(4) None of these

42) A body is thrown vertically up with certain initial velocity, the potential and kinetic energies of the body are equal at a point P in its path. If the same body is thrown with double the velocity upwards, the ratio of potential and kinetic energies of the body when it crosses the same point, is :

(1) 1:1

(2) 1:4

(3) 1:7

(4) 1:8

43) Power applied to a particle varies with time as $P = (3t^2 - 2t + 1)$ watts. Where t is time in seconds. Then the change in kinetic energy of particle between time t = 2s to t = 4s is :-

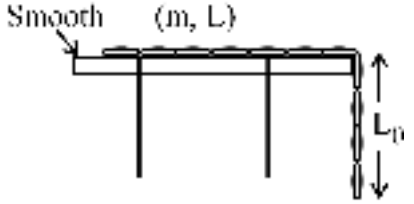
(1) 46 J

(2) 52 J

(3) 92 J

(4) 104 J

44) Consider a uniform chain of mass m and length L, the chain is released when hanging length is



0

L . Find the speed with which the chain leaves the table :-

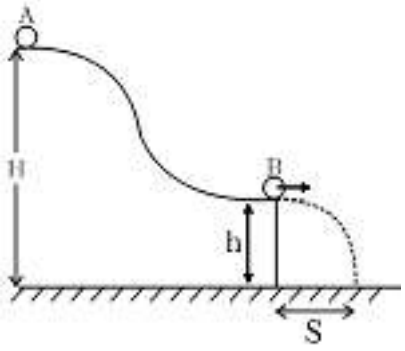
(1) $\sqrt{\frac{g}{L}(L^2 - L_0^2)}$

(2) $\sqrt{\frac{g}{L}(L_0^2 - L^2)}$

(3) $\sqrt{\frac{2g}{L}(L^2 - L_0^2)}$

(4) $\sqrt{\frac{g}{2L}(L^2 - L_0^2)}$

45) A small disc A slides down with initial velocity equal to zero from the top of a smooth hill of height H having a horizontal portion. The height of the horizontal portion is h to ensure that the



distance s covered by the disc is maximum, the values of h and S are?

(1) $h = \frac{H}{2},\ s = \frac{H}{2}$

(2) h = H, s = 2H

(3) $h = \frac{H}{2},\ s = H$

(4) h = 2H, s = H

CHEMISTRY

1) What is the molar solubility of $Fe(OH)_2$
($K_{sp} = 8.0 \times 10^{-16}$) at pH 13.0 ?

(1) 8.0×10^{-18}

(2) 8.0×10^{-15}

(3) 8.0×10^{-17}

(4) 8.0×10^{-14}

2) In the following

$$\underset{(A)}{[Al(H_2O)_6]^{+3}} + \underset{(B)}{HCO_3^-} \rightleftharpoons \underset{(C)}{[Al(H_2O)_5OH]^{+2}} + \underset{(D)}{H_2CO_3}$$

Species behaving as Bronsted – lowry acid are:-

(1) (A), (D)

(2) (A), (C)

(3) (B), (C)

(4) (B), (D)

3) Given K_a values of 5.76×10^{-10} and 4.8×10^{-10} for NH_4^+ and HCN respectively. What is the equilibrium constant for the following reaction ?
$NH_4^+(aq.) + CN^-(aq.) \rightleftharpoons NH_3(aq.) + HCN(aq.)$

(1) 0.83

(2) 1.2

(3) 8.0×10^{-11}

(4) 27.6×10^{-10}

4) Which of the following will produce a buffer solution when mixed in equal volumes?

(1) 0.1 mol dm^{-3} NH_4OH and 0.1 mol dm^{-3} HCl

(2) 0.05 mol dm^{-3} NH_4OH and 0.1 mol dm^{-3} HCl

(3) 0.1 mol dm^{-3} NH_4OH and 0.05 mol dm^{-3} HCl

(4) 0.1 mol dm^{-3} CH_3COONa and 0.1 mol dm^{-3} NaOH

5) What is the $[OH^-]$ in the final solution prepared by mixing 20.0 mL of 0.050 M HCl with 30.0 mL of 0.10M $Ba(OH)_2$?

(1) 0.10 M

(2) 0.12 M

(3) 0.40 M

(4) 0.0050M

6) If a solution has 100 times as many hydroxide ions as in pure water at 25°C, then the pH of the solution at the same temperature will be

(1) 9

(2) 6.4

(3) 7

(4) 13.2

7) $Hg_{(\square)} + Cl_{2(g)} \rightarrow HgCl_{2(s)}$; $\Delta H° = -230$ kJ/mol
$Hg_2Cl_{2(s)} + Cl_{2(g)} \rightarrow 2HgCl_{2(s)}$; $\Delta H° = -300$ kJ/mol
What is the $\Delta H°$ for the reaction,
$2Hg(\square) + Cl_2 \rightarrow Hg_2Cl_{2(s)}$

(1) –5.60 kJ/mol

(2) –70 kJ/mol

(3) –160 kJ/mol

(4) +160 kJ/mol

8) Given the following bond energies :-
C–H = 414 kJ/mol C–Cl = 150 kJ/mol
Cl–Cl = 243 kJ/mol H–Cl = 432 kJ/mol
How much energy would be required for the reaction,
$CH_{4(g)} + 2Cl_{2(g)} \rightarrow CH_2Cl_{2(g)} + 2HCl_{(g)}$

(1) 150 kJ

(2) 130 kJ

(3) 228 kJ

(4) 571 kJ

9) The enthalpies of neutralization of a weak base AOH and a strong base BOH by HCl are –12250 cal/mol and –13000 cal/mol respectively. When one mole of HCl is added to a solution containing 1 mole of AOH and 1 mole of BOH, the enthalpy change was – 12500 cal/mol. In what ratio is the acid

distribution between AOH and BOH ?

(1) 2 : 1
(2) 2 : 3
(3) 1 : 2
(4) None of these

10) Calorific value of hydrogen gas is 143 kJg^{-1}. The standard enthalpy of formation of H_2O will be -

(1) – 143 kJ mol^{-1}
(2) + 143 kJ mol^{-1}
(3) – 286 kJ mol^{-1}
(4) + 286 kJ mol^{-1}

11) Which of the following is true concerning the entropy of the universe for a chemical reaction that has reached equilibrium?

(1) The entropy is constantly changing.
(2) The entropy has reached a maximum value
(3) The entropy has reached a minimum value
(4) The system has reached zero entropy.

12) One monoatomic gas is expanded adibatically from 2L to 10 L at against constant 1 atm external pressure find ΔU (in atm L) ?
[1 L - atm = 101J]

(1) –8
(2) 0
(3) –808
(4) 58.2

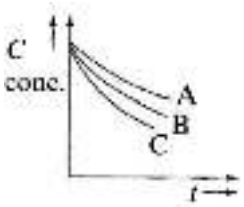
13) At 27°C, two moles of an ideal gas undergoes a change in volume from 10 L to 100 L isothermally. Then calculate the maximum work done during the process?

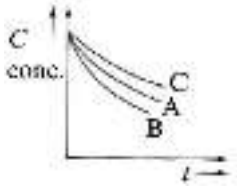
(1) –11.5 kJ
(2) –23 kJ
(3) –5.7 kJ
(4) None of these

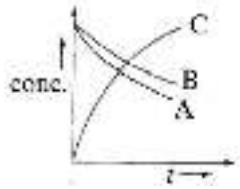
14) For a reversible process -

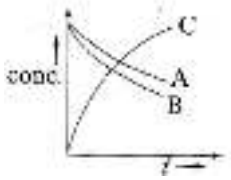
(1) System & surrounding are in equilibrium only at initial & final state.
(2) $P_{int} = P_{ext} \pm \Delta P$; P_{ext} is constant
(3) Process complete in short time
(4) Driving force should be infinitesimally greater than opposing force

15) Which of the following graphs is correct representation for the reaction of type: $A + 2B \rightarrow C$?

(1) 

(2) 

(3) 

(4) 

16)

The charge on 1 gm-ions of Al^{3+} is: (N_A = Avogadro number, e = charge on one electron)

(1) $\frac{1}{27} N_A$ e coulomb

(2) $\frac{1}{3} \times N_A$ e coulomb

(3) $\frac{1}{9} \times N_A$ e coulomb

(4) $3 \times N_A$ e coulomb

17) Number of Oxygen molecules having weight equal to weight of 20 molecules of SO_3.

(1) 100

(2) 50

(3) 15

(4) 8

18) An organic substance containing carbon, hydrogen and oxygen gave the following percentage composition.

C = 40.7%; H = 5.1% and O = 54.2%

Calculate the empirical formula weight of the compound. (At.wt of C = 12, O = 16, H = 1)

(1) 118

(2) 59

(3) 177

(4) 133

19) Chlorine is prepared according to the reaction :-
$4HCl + MnO_2 \rightarrow 2H_2O + MnCl_2 + Cl_2$
How many grams of HCl react with 5g of manganese dioxide ? (Mn = 55)

(1) 8.39 gm

(2) 6.76 gm

(3) 9.72 gm

(4) 10 gm

20) Rearrange the following (I to IV) in the order of increasing masses :
(I) 0.5 mole of O_3
(II) 0.5 gm atom of oxygen
(III) 3.011×10^{23} molecules of O_2
(IV) 5.6 litre of CO_2 at STP

(1) II < IV < III < I

(2) II < I < IV < III

(3) IV < II < III < I

(4) I < II < III < IV

21) For reaction
$2A + 3B \rightarrow 5C$
$4C + 7D \rightarrow 9E + 6F$
$3F + 4G \rightarrow 8H$
Calculate maximum number of mole of H, which can be produce by 5 mole of A and 10 mole of B :-

(1) 50

(2) 50/3

(3) 10

(4) 10/3

22) **Assertion :** In CuO equivalent weight of Cu is 63.5 & in Cu_2O equivalent weight of Cu is 31.8.
Reason : Equivalent weight of any metal is the quantity of metal in gram which is combined with 16 g of oxygen in the formation of metal oxide.

(1) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.

(2) Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.

(3) Assertion is true but Reason is false.

(4) Both Assertion and Reason are false.

23) For the reaction $N_2O_4 \rightleftharpoons 2NO_2(g)$at 77 °C, the value of K_C = 0.4 mol/L. The value of K_P for the reaction at the same temperature would be:

(1) 11.48 atm

(2) 1.148 atm

(3) 1.2×10^{-2} atm

(4) 1.4×10^{-3} atm

24) In which reaction will an increase in the volume of the container favour the formation of products?

(1) $C(s) + H_2O(g) \rightleftharpoons CO(g) + H_2(g)$

(2) $H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$

(3) $4NH_3(g) + 5O_2(g) \rightleftharpoons 4NO(g) + 6H_2O(\ell)$

(4) $3O_2(g) \rightleftharpoons 2O_3(g)$

25) For the following reactions, equilibrium constants are given :
$S(s) + O_2(g) \rightleftharpoons SO_2(g)$; $K_1 = 10^{52}$
$2S(s) + 3O_2(g) \rightleftharpoons 2SO_3(g)$; $K_2 = 10^{129}$
The equilibrium constant for the reaction,
$2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$ is :

(1) 10^{181}

(2) 10^{154}

(3) 10^{25}

(4) 10^{77}

26) 2 mole each of SO_3, CO, SO_2 and CO_2 is taken in a one lit. vessel. If K_C for
$SO_3(g) + CO(g) \rightleftharpoons SO_2(g) + CO_2(g)$ is 1/9 then (n = mole)

(1) total no. of moles at equilibrium are less than 8

(2) $n(SO_3) + n(CO_2) = 4$

(3) $[n(SO_2)/n(CO)] < 1$

(4) both (2) and (3)

27) $x \rightleftharpoons 2y \qquad K_P$
$A \rightleftharpoons B + C \quad K_P'$

If degree of dissociation of x and A are same and $K_P = 2K_P'$ then the ratio of total pressure $\frac{P}{P'}$ = ?

(1) $\frac{1}{2}$

(2) $\frac{1}{3}$

(3) $\frac{1}{4}$

(4) 2

28) In a reversible reaction with equilibrium constant = 25, the rate of forward reaction is 5 times faster than that of reverse reaction. What is the value of reaction quotient.

(1) 2.5

(2) 5

(3) 7

(4) 10

29) A + 3B ⇌ 2C If initially 1 mole of A & B each were taken in 10L container then select correct relation at equilibrium :-

(1) [A] > [B]

(2) [B] > [A]

(3) [A] = [B]

(4) [B] = 2[A]

30) Lattice enthalpy of NaCl(s) is 788 kJ/mol. Enthalpy of hydration of $Na^+(g)$ and $Cl^-(g)$ is –300 kJ/mol and –484 kJ/mol respectively . Then heat of solution ($\Delta_{sol}H$) of NaCl(s) is :

(1) –4 kJ/mol

(2) 12 kJ/mol

(3) 4 kJ/mol

(4) 0

31) Which of following is not a redox change ?

(1) $2H_2S + SO_2 \rightarrow 2H_2O + 3S$

(2) $2BaO + O_2 \rightarrow 2BaO_2$

(3) $BaO_2 + H_2SO_4 \rightarrow BaSO_4 + H_2O_2$

(4) $2KClO_3 \rightarrow 2KCl + 3O_2$

32) $xMnO_4^- + yCN^- + wH_2O \rightarrow xMnO_2 + yCNO^- + zOH^-$

in the balanced chemical reaction x : y : z is :

(1) 2 : 3 : 1

(2) 1 : 3 : 2

(3) 2 : 1 : 3

(4) 2 : 3 : 2

33) The total number of millimoles of NaOH required to completely react with 20 ml of 0.1 M NaH_2PO_3 solution and 60 mL of 0.1 M $NaHCO_3$:

(1) 8

(2) 6

(3) 3

(4) 2

34) $KMnO_4$ reacts with oxalic acid according to the equation :-

$2MnO_4^- + 5C_2O_4^{2-} + 16H^+ \rightarrow 2Mn^{+2} + 10CO_2 + 8H_2O$
Here, 20 mL of 0.1 M $KMnO_4$ is equivalent to:-

(1) 120 mL of 0.25 M $H_2C_2O_4$

(2) 150 mL of 0.10 M $H_2C_2O_4$

(3) 25 mL of 0.20 M $H_2C_2O_4$

(4) 50 mL of 0.20 M $H_2C_2O_4$

35) (i) $4NH_3 + 5O_2 \rightarrow 4NO + 6H_2O$,
(ii) $2NH_3 \rightarrow N_2 + 3H_2$ The ratio of equivalent masses of NH_3 in the reactions (i) and (ii) are :-

(1) 5 : 6

(2) 6 : 5

(3) 5 : 3

(4) 3 : 5

36)

When CrI_3 oxidises to $Cr_2O_7^{-2}$ and IO_4^- equivalent mass of CrI_3 will be :-

(1) $\frac{M}{33}$

(2) $\frac{M}{27}$

(3) $\frac{M}{28}$

(4) $\frac{M}{24}$

37) If x mole of $FeSO_4$ require y mole of $KMnO_4$ for complete oxidation in acidic medium then how many mole of $KMnO_4$ will completely oxidise x mole of FeC_2O_4 in acidic medium :-

(1) y

(2) $\frac{y}{2}$

(3) 3y

(4) $\frac{y}{3}$

38) **Assertion (A) :** Boiling point of liquid depends on the altitude of the place, at high altitude the boiling point decreases.
Reason (R) : Boiling point of liquid decreases as external pressure on liquid decreases.

(1) A and R both are correct and R is correct explanation of A

(2) A and R both are correct and R is not correct explanation of A.

(3) A is correct but R is incorrect

(4) A is incorrect but R is correct

39) Consider the following statements about a volatile liquid present in an open vessel.
(A) The rate of evaporation remains constant.
(B) The rate of condensation from vapour to liquid state is greater than the rate of evaporation.
(C) It is not possible to reach liquid and vapour in equilibrium in an open system.
Correct statements are/is :-

(1) A, B and C

(2) A and B only

(3) A and C only

(4) B and C only

40) Match the column-I with column-II

Column-I		**Column-II**	
(P)	Conversion of water into steam at 1 atm and 400 K	(I)	$\Delta G = 0$
(Q)	Isothermal compression of an ideal gas	(II)	$\Delta G > 0$
(R)	Conversion of ice into water at 1 atm and 273 K	(III)	$\Delta S = 0$
(S)	Heating of a liquid reversibly adiabatically	(IV)	$\Delta G < 0$

(1) P → IV ; Q → III ; R → II ; S → I

(2) P → III ; Q → I ; R → II ; S → IV

(3) P → II ; Q → IV ; R → I ; S → III

(4) P → IV ; Q → II ; R → I ; S → III

41) 100 mL, 0.1 M of weak base BOH is titrated with 0.1 M of HCl solution. Match column-I with column-II. (ionisation constant of BOH = K_b)

Column-I (Volume of HCl added)		**Column-II (pH of resultant solution)**	
(P)	25 mL	(I)	$pH = pK_w - pK_b - \log 3$
(Q)	50 mL	(II)	$pH = \frac{1}{2}(pK_w - pK_b + \log 20)$
(R)	75 mL	(III)	$pH = pK_w - pK_b + \log 3$
(S)	100 mL	(IV)	$pH = pK_w - pK_b$

(1) P → IV ; Q → III ; R → II ; S → I

(2) P → III ; Q → IV ; R → I ; S → II

(3) P → I ; Q → II ; R → III ; S → IV

(4) P → II ; Q → I ; R → IV ; S → III

42) Consider the following data for 0.01 M NH_4Cl aqueous solution at 25°C.

(A) $[Cl^-_{(aq)}] < 10^{-2}$ M
(B) $[NH_4^+{}_{(aq)}] < 10^{-2}$ M
(C) pOH > 7
(D) $[H^+] > 10^{-7}$ M
Correct options from above are/is :-

(1) Only A
(2) A, B, C and D
(3) Only B, C and D
(4) Only A and D

43) **Assertion (A) :** Phenolphthalein is suitable indicator for the titration of weak acid and strong base.
Reason (R) : In titration of weak acid and strong base, at equivalence point pH of solution is nearly in the range 8.2 to 10.2.

(1) Both A and R are correct and R is correct explanation of A.
(2) Both A and R are correct and R is not correct explanation of A.
(3) A is correct but R is not correct
(4) A is not correct but R is correct

44) **Statement-I :** At constant temperature on dilution solubility of $AgCl_{(s)}$ in water increases.
Statement-II : On dilution moles of AgCl dissolved in water increases.

(1) Both statement-I and II are correct
(2) Both statement-I and II are incorrect
(3) Statement-I is correct but statement-II is incorrect
(4) Statement-I is incorrect but statement-II is correct

45) **Assertion (A) :** When H_3PO_2 dispropotionate into PH_3 and H_3PO_3 , equivalent mass of H_3PO_2 is 49.5.
Reason (R) : Valency factor of H_3PO_2 is 4/3.

(1) Both A and R are correct and R is correct explanation of A
(2) Both A and R are correct and R is not correct explanation of A
(3) A is correct but R is incorrect
(4) A is incorrect but R is correct

BIOLOGY

1) For unicellular organisms reproduction is synonymous with _____.

(1) Growth
(2) Cellular organisation
(3) Consciousness
(4) Metabolism

2) The family of *Mangifera indica*, *Triticum aestivum* and *Solanum nigrum* are respectively :-

(1) Poaceae, Solanaceae and Anacardiaceae.
(2) Anacardiaceae, Poaceae and Solanaceae.
(3) Anacardiaceae, Felidae and Poaceae.
(4) Solanaceae, Canidae and Felidae.

3) Artificial system of classification classifies plants on the basis of ?

(1) One or two morphological characters
(2) Phylogenetic trends
(3) Many naturally existing characters
(4) All morphological characters

4) **Statement-I :** Sexual cycle of fungi involves plasmogamy, karyogamy and meiosis.
Statement-II : Members of deuteromycetes reproduce by asexual spores.

(1) Both statement I and II are correct.
(2) Both statement I and II are incorrect.
(3) Statement I is correct but statement II is incorrect.
(4) Statement I is incorrect but statement II is correct.

5) _A_ bacteria are most abundant in nature. Some time under _B_ conditions, bacteria produce spores. A and B are :-

(1) A- Autotrophic, B- Favourable
(2) A- Autotrophic, B- Unfavourable
(3) A- Heterotrophic, B- Unfavourable
(4) A- Heterotrophic, B- Favourable

6) Mycoplasma belongs to kingdom _A_, they can survive in the absence of _B_.

(1) A-Monera, B-Oxygen
(2) A-Protista, B-CO_2
(3) A-Fungi, B-Oxygen
(4) A-Fungi, B-CO_2

7) Read the following characters :-
Cell type - Eukaryotic
Cell wall - Present in some
Nuclear membrane - Present
Body Organisation - Cellular
Mode of nutrition - Autotrophic (Photosynthetic) and heterotrophic
Characters given above are related to :-

(1) Kingdom Monera and Protista
(2) Only Kingdom Protista

(3) Only Kingdom Fungi

(4) Kingdom Monera and Plantae

8) **Statement-I :** Asexual spore are generally absent in basidiomycetes.
Statement-II : Sex organ are absent in basidiomycetes.

(1) Both **Statement I** and **Statement II** are incorrect.

(2) **Statement I** is correct but **Statement II** is incorrect.

(3) **Statement I** is incorrect but **Statement II** is correct.

(4) Both **Statement I** and **Statement II** are correct.

9) Which of the following help in DNA replication in bacterial cell ?

(1) Lipopolysaccharide wall

(2) Peptidoglycan wall

(3) Mesosome

(4) Ribosome

10) Storage bodies-pyrenoides in the members of chlorophyceae are located in :-

(1) Mitochondria

(2) Cell membrane

(3) Chloroplast

(4) Smooth endoplasmic reticulum

11) Isogamous, anisogamous and oogamous all three types of sexual reproduction is found in which class of algae?

(1) Chlorophyceae

(2) Rhodophyceae

(3) Cyanophyceae

(4) Myxophyceae

12) Algin and carrageen are hydrocolloids which is produce respectively by :-

(1) Rhodophyceae and phaeophyceae

(2) Phaephyceae chlorophyceae

(3) Chlorophyceae and rhodophyceae

(4) Phaeophyceae and Rhodophyceae

13) Identify the given figure : 

(1) *Adiantum*
(2) *Pteris*
(3) *Dryopteris*
(4) *Salvinia*

14) Ploidy of the protonemal cell of moss is :-

(1) Haploid
(2) Diploid
(3) Triploid
(4) Polyploid

15) Roots of gymnosperms (Pinus) have fungal association in the form of :-

(1) Lichen
(2) Coralloid roots
(3) Rhizome
(4) Mycorrhiza

16) Which of the following statements are true ?
(a) Vast majority of bacteria are autotrophs.
(b) Bacteria reproduce mainly by fission.
(c) Mycoplasma are pathogenic to plants only.
(d) All cyanobacteria can fix atmospheric nitrogen.
Options :

(1) a, b and c only
(2) a, c and d only
(3) b only
(4) b, c only

17) Which of the following statement is incorrect about phycomycetes ?

(1) Member of phycomycetes are obligate parasites on plants.
(2) The mycelium of phycomycetes is septate and coenocytic.
(3) Asexual spores are endogenously produced in sporangium.
(4) *Albugo* is the parasitic fungi on mustard.

18) Consider the following characters
(i) Heterotropic organisms
(ii) A dikaryon stage
(iii) Cell wall made up of chitin
(iv) Zygotic meiosis division
Which of the above statement related to fungi ?

(1) (i), (ii), (iii) only
(2) (i), (iii), (iv) only
(3) (ii), (iii), (iv) only
(4) (i), (ii), (iii), (iv)

19) Which one of the following statements is incorrect ?

(1) Members of protista are primarily aquatic.
(2) Protistan cell body contains a well defined nucleus.
(3) Protists reproduce both asexually and sexually.
(4) All protistan cell body have flagella or cillia.

20) Select correct & incorrect statements :-
(i) Chemosynthetic bacteria oxidise various inorganic substances.
(ii) Chemosynthetic bacteria play great role in recycling nutrients.
(iii) Colonies of BGA generally remain surrounded by gelatinous sheath.
(iv) Chlorophyll-a pigments of cyanobacteria are similar to green plants.

(1) Correct statement-1, incorrect statement-3
(2) Correct statement-2, incorrect statement-2
(3) Correct statement-3, incorrect statement-1
(4) All statements are correct.

21) **Statement-I** : Membranous extensions into the cytoplasm called chromatophores which contain pigments are found in all prokaryotes.
Statement-II : Pili and fimbriae are surface structures of the bacteria and play a role in motility.

(1) Both statement-I and statement-II are correct
(2) Both statement-I and statement-II are incorrect
(3) Statement-I is correct but statement-II is incorrect
(4) Statement-I is incorrect but statement-II is correct

22) **Statement-I:** In addition to proteins, viruses also contain genetic material that would be either RNA or DNA
Statement-II: Some virus contain both RNA and DNA.

(1) Both statement are correct
(2) Only statement I is correct
(3) Only statement II is correct
(4) Both statement are incorrect

23) **Statement-I :** Linnaeus system of classification distinguish between the eukaryotes and prokaryotes.
Statement-II : The kingdom Monera is divided into three domains.

(1) Both statement-I & statement-II are correct

(2) Both statement-I & statement-II are incorrect

(3) Statement-I is correct but statement-II is incorrect

(4) Statement-I is incorrect but statement-II is correct

24) Match the column I with column II and choose the correct option -

	Column-I		**Column-II**
A	*Anabaena*	i	Independent gametophyte
B	*Cycas*	ii	Dioecious sporophyte
C	*Funaria*	iii	Pyriform gametes
D	*Ectocarpus*	iv	Nitrogen fixation

Option :

(1) A - i, B - iii, C - iv, D - ii

(2) A - iii, B - i, C - ii, D - iv

(3) A - iv, B - i, C - ii, D - iii

(4) A - iv, B - ii, C - i, D - iii

25)

Match List-I with List-II

	List-I		**List-II**
A	Gemma cup	i	*Sphagnum*
B	Peat Moss	ii	*Funaria* and *Polytrichum*
C	*Mosses*	iii	*Marchantia*
D	Capsule	iv	Part of sporophyte in bryophytes

Choose the correct answer from the option given below :

	A	B	C	D
1	iii	i	ii	iv
2	ii	iii	iv	i
3	ii	iii	i	iv
4	iii	iv	ii	i

(1) 1

(2) 2

(3) 3

(4) 4

26) Match the column-I with column-II

(A)	Anisogamous	(i)	*Ulothrix*
(B)	Oogamous	(ii)	*Eudorina*
(C)	Flagellated gametes	(iii)	*Spirogyra*
(D)	Non-flagellated gamets	(iv)	*Volvox*

	A	B	C	D
(1)	iv	ii	ii	iii
(2)	i	ii	iii	iv
(3)	ii	iv	i	iii
(4)	ii	iv	iii	i

(1) 1

(2) 2

(3) 3

(4) 4

27) Match list-I with list-II :-

	List-I		**List-II**
(A)	Brown algae	(I)	*Cedrus*
(B)	Liverwort	(II)	*Pteris*
(C)	Fern	(III)	*Marchantia*
(D)	Gymnosperm	(IV)	*Fucus*

Choose correct answer from the options given below :-

(1) A-II, B-III, C-I, D-IV

(2) A-IV, B-III, C-II, D-I

(3) A-IV, B-II, C-III, D-I

(4) A-IV, B-II, C-I, D-III

28) Given below are two statements : One is labelled as assertion (A) and the other is labelled as reason (R).
Assertion (A) :- Sexual reproduction in *Spirogyra* is isogamous.
Reason (R) :- Gametes in *Spirogyra* are flagellated and similar in size.
Choose the correct answer from the options given below :-

(1) Both (A) and (R) are true and (R) is correct explanation of (A).

(2) Both (A) and (R) are true but (R) is not correct explanation of (A).

(3) (A) is true but (R) is false.

(4) (A) is false but (R) is true.

29) Given below are two statements :- One is labelled a Assertion (A) and the other in labelled as Reason (R).
Assertion (A) :- The protonema is a creeping, green, branched and frequently filamentous stage.
Reason (R) :- In mosses, protonema stage bears the sex organs.
In the light of above statements, choose the most appropriate answer from the options given below.

(1) Both A & R are correct but R is not the correct explanation of A.

(2) A is correct but R is not correct.

(3) A is not correct but R is correct

(4) Both A & R are correct and R is the correct explanation of A.

30) **Assertion (A) :** In gymnosperms the female gametophytes do not have an independent free living existance.
Reason (R) : In gymnosperms female gametophytes remain within the sporangia retained on sporophyte.

(1) Both (A) and (R) are correct and (R) is the correct explanation of (A).

(2) (A) is correct but (R) is incorrect.

(3) (A) is incorrect but (R) is correct.

(4) Both (A) and (R) are correct but (R) is not the correct explanation of (A).

31) **Assertion (A)** - The problem of classification becomes more complex at higher taxonomic categories.
Reason (R) - As we go from species to Kingdom, the number of common characteristics goes on decreasing.

(1) Both (A) and (R) are correct and (R) is correct explanation of (A)

(2) Both (A) and (R) are correct but (R) is not correct explanation of (A)

(3) (A) is correct but (R) is incorrect

(4) (A) is incorrect but (R) is correct

32) Match the Column-I and Column-II and choose the correct option-

	Column-I		**Column-II**
(i)	Man	(a)	*Musca domestica*
(ii)	House fly	(b)	*Mangifera indica*
(iii)	Mango	(c)	*Triticum aestivum*
(iv)	Wheat	(d)	*Homo sapiens*

(1) (i–d), (ii–a), (iii–b), (iv–c)

(2) (i–a), (ii–d), (iii–b), (iv–c)

(3) (i–d), (ii–b), (iii–a), (iv–c)

(4) (i–d), (ii–a), (iii–c), (iv–b)

33) **Statement-I :** Classification is not a single step process but involves hierarchy of steps.
Statement-II: The taxonomic categories are distinct biological entities and not merely

morphological aggregates.

(1) Both **Statement I** and **Statement II** are correct.

(2) **Statement I** is correct but **Statement II** is incorrect.

(3) **Statement I** is incorrect but **Statement II** is correct.

(4) Both **Statement I** and **Statement II** are incorrect.

34) How many given below statements are correct statements ?
(A) Diptera is class of Housefly.
(B) Hominidae is a order of Man.
(C) Monocotyledonae is a class of wheat
(D) Mangifera is a genus of Mango.
(E) Musca is a genus of housefly.

(1) Two

(2) Three

(3) Four

(4) Five

35) In lichens, the algal component is known as ___A___ and fungal component as ___B____, which are ___C___ and ____D____ respectively.

(1) A-Mycobiont, B-Phycobiont, C-Autotrophic, D-Heterotrophic

(2) A-Phycobiont, B-Mycobiont, C-Heterotrophic, D-Autotrophic

(3) A-Mycobiont, B-Phycobiont, C-Heterotrophic, D-Autotrophic

(4) A-Phycobiont, B-Mycobiont, C-Autotrophic, D-Heterotrophic

36) How many of the following in the given list are placed under kingdom plantae according to five-kingdom classification system ?
Bacteria, Blue green algae, *Chlamydomonas*, *Chlorella*, *Agaricus*, Mosses, Ferns, *Amoeba*, Gymnosperms, Angiosperms, *Paramoecium*.
Options :

(1) Nine

(2) Seven

(3) Eight

(4) Four

37) I. ___A___ commonly known as imperfect fungi because only the asexual or vegetative phase of these fungi are known.
II. The mycelium of bracket fungi is ____B____ and ___C____.
III. The asexual spore of sac fungi are conidia produced ____D_____ on the special mycelium.
Identify the A, B, C and D -

	A	B	C	D
1.	Ascomycetes	Septate	Unbranched	Exogenously
2.	Deuteromycetes	Septate	Branched	Exogenously

3.	Basidiomycetes	Aseptate	Unbranched	Endogenously
4.	Deuteromycetes	Aseptate	Unbranched	Endogenously

(1) 1

(2) 2

(3) 3

(4) 4

38) Study the following statements with respect to Bentham and Hooker classification:-
(A) It is largest and natural classification of Spermatophyta.
(B) It was mainly based on floral characters.
(C) It is a simplest classification because it is based on actual observation of plants.
(D) In this classification phylogeny is not considered.
How many statement is/are correct?

(1) One

(2) Two

(3) Three

(4) Four

39) Why mycoplasma kept under monera kingdom ?
(a) smallest organism
(b) membrane bound cell organells absent
(c) cell wall absent
(d) Nucleus absent
(e) 70 s ribosome present
(f) Histone protein absent in DNA
Choose the most appropriate answer from the option given below :-

(1) (a), (b) only

(2) (e), (f) only

(3) (b), (c), (d) only

(4) (a), (e), (f) only

40) Match the column A with B :-

Column - A		**Column - B**	
(a)	*Nostoc*	(i)	Heterotroph
(b)	Curd making bacteria	(ii)	Pathogen
(c)	*Mycoplasma*	(iii)	Heterocyst
(d)	Methanogen	(iv)	Biogas

(1) a–i, b–ii, c–iii, d–iv

(2) a–iii, b–i, c–ii, d–iv

(3) a–ii, b–i, c–iv, d–iii

(4) a–i, b–iii, c–ii, d–iv

41) Which of the following pairs with respect to **Cycas** are correctly matched :
(A) Megasporophyll = 2n
(B) Microsporangium = n
(C) Female gametophyte = n
(D) Endosperm = n
(E) Female cone = 2n

(1) A, C, D, E only
(2) A, B, C, D, E
(3) A, C, D only
(4) A, C, E only

42) How many of the following statements are correct :-
(A) Agar, one of the commercial product obtained from *Gelidium* and *Gracilaria*.
(B) Female sex organ in bryophytes is archegonium which is flask shaped and produces many eggs.
(C) Evolutionarily pteridophytes are first terrestrial plants to possess vascular tissues.
(D) The stems are unbranched in *Cycas* and branched in Pinus, *Cedrus*
(E) In angiosperms, the seeds are enclosed in fruits.

(1) All statements are correct.
(2) (A), (B), (C) and (D) are correct.
(3) All statements are correct except (B)
(4) (A), (B), (D), (E) are correct.

43) Identify the given diagrams (a), (b), (c) and (d).



(1) (a) *Salvinia* ; (b) *Selaginella* ; (c) *Ferns* ; (d) *Equisetum*
(2) (a) *Selaginella* ; (b) *Equisetum* ; (c) *Salvinia* ; (d) *Ferns*
(3) (a) *Selaginella* ; (b) *Equisetum* ; (c) *Ferns* ; (d) *Salvinia*
(4) (a) *Equisetum* ; (b) *Ferns* ; (c) *Selaginella* ; (d) *Salvinia*

44) Fill in the blanks with correct option:-
(i) The plant body of bryophytes is _____(a)_____ differentiated than that of algae.

(ii) The sex organs of bryophytes are _____(b)_____.
(iii) The first stage of gametophyte in mosses is _____(c)_____, which develops directly from a _____(d)_____.
(iv) The sporophyte in mosses is _____(e)_____ elaborate than that in liverworts.

(1) (a) more, (b) multicellular, (c) protonema, (d) spore, (e) more
(2) (a) less, (b) multicellular, (c) protonema, (d) spore, (e) less
(3) (a) more, (b) multicellular, (c) prothallus, (d) spore, (e) more
(4) (a) more, (b) multicellular, (c) prothallus, (d) zygote, (e) less

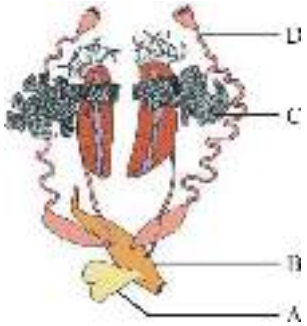
45) How many of the following members belong to Phaeophyceae ?
Chara, *Ectocarpus*, *Porphyra*, *Dictyota*, *Gelidium*, *Sargassum*, *Fucus*
Choose the correct option :

(1) Three
(2) Two
(3) Four
(4) Five

46) Which of the following statement is not true :
(a) The vascular system of frog is well developed closed type and also a lymphatic system is present
(b) Vascular system consist of blood vessels, heart and blood
(c) While lymphatic system consist of lymph channels, lymph and lymph nodes
(d) Heart is a muscular structure present in upper part of body

(1) a and c are not true
(2) All are true
(3) d not true
(4) c not true

47) Given below is a diagramatic representation of female reproductive system of frog labelled A, B, C and D :



Find out incorrect one :-

(1) There is functional connection between C and kidney.
(2) B is small, median chamber that is used to pass faecal matter, urine and ova to the exterior.
(3) D is one pair arising from the ovaries opens into the cloaca separately.
(4) A is thin walled urinary bladder is present ventral to rectum which also opens in cloaea.

48) **Assertion :** Frogs exhibit sexual dimorphism.
Reason : Male frog can be distinguished by the presence of sound producing vocal sacs which is absent in female frogs.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

49) **Statement-I :** In frog each kidney is composed of several structural and functional units called uriniferous tubules.
Statement-II : In frog a triangular structure called sinus venosus joins the left atrium.

(1) Both statement-I and statement-II are incorrect.

(2) statement-I is correct but statement-II is incorrect

(3) statement-I is incorrect but statement-II is correct

(4) Both statement-I and statement-II are correct

50) Select the incorrect statement w.r.t frog :-

(1) On land, frog respire only through lungs

(2) Digestion in frog begins in stomach

(3) During summer sleep frog respire through skin

(4) Tadpole larva has gills for respiration.

51) Match the column in refrence to frog :-

Column-I		**Column-II**	
A.	Cutaneous respiration	(i)	Receive sound signals
B.	Tympanum	(ii)	1st digit of fore-limbs
C.	Copulatory pad	(iii)	Aquatic Respiration
D.	Ureotelic	(iv)	Excrete urea

(1) A–i, B–ii, C–iii, D–iv

(2) A–ii, B–i, C–iii, D–iv

(3) A–iii, B–i, C–ii, D–iv

(4) A–iii, B–i, C–iv, D–ii

52) In forelimb and hindlimbs of frogs how many digits are present respectively :-

(1) Five and four

(2) Four and five

(3) Five and five

(4) Four and four

53) Tadpole larva of frog excretes ....(a).... while adult frog excretes ....(b)...., a and b in given statement are -

(1) (a) Ammonia ; (b) Ammonia

(2) (a) Ammonia ; (b) Urea

(3) (a) Urea ; (b) Uric acid

(4) (a) NH_3 ; (b) Guanine

54) Sinus venosus in the heart of frog joins the

(1) Left atrium

(2) Ventricle

(3) Truncus arteriosus

(4) Right atrium

55)

How many ova laying at a time by mature female frog ?

(1) 2500-3000

(2) 3000-4000

(3) 4000-5000

(4) 6000-7000

56) Find out the **correct** option with respect to location of different structure in cockroach :-

	Anal style	**Mushroom gland in male**	**Anal cerci**	**Sperma thecae in female**
(1)	10th Sterna	2-6 Abdominal segment	9th Terga	4-6th Abdominal segment
(2)	9th Sternum	6-7 Abdominal segment	10th Terga	6th Abdominal segment
(3)	7th Sterna	2-6 Abdominal segment	8th Sterna	4-6th Abdominal segment
(4)	10th Terga	6-7 Abdominal segment	9th Sterna	2-6th Abdominal segment

(1) 1

(2) 2

(3) 3

(4) 4

57) Consider the following four statements (i) - (iv) and select the correct option stating which ones are true (T) and which ones are false (F).
(i) In male cockroach, genital pouch or chamber lies at the hind end of abdomen bounded dorsally by 9th and 10th terga and ventrally by the 9th sternum.

(ii) In cockroach, the haemolymph is composed of colourless plasma and haemocytes.
(iii) In female cockroach each ovary is formed of a group of ten ovarian tubules or ovarioles, containing a chain of developing ova.
(iv) In cockroach, the nymph grows by moulting about 13 times to reach the adult form.

(1) (i)-F, (ii)-T, (iii)-F, (iv)-T
(2) (i)-F, (ii)-F, (iii)-T, (iv)-T
(3) (i)-T, (ii)-T, (iii)-F, (iv)-T
(4) (i)-T, (ii)-F, (iii)-T, (iv)-F

58) Given below is a diagrammatic representation of alimentary canal of cockroach.
Choose the option which is not correct with respect to the function of labelled structure A, B, C and D



(1) Oesophagus opens into a sac like structure labelled as A used for storing of food.
(2) B helps in grinding of food because it have an outer thick cuticular layer and inner muscular layer.
(3) C is present at the junction of foregut and midgut and secretes digestive juice.
(4) D helps in removal of excretory product from haemolymph.

59) Select the incorrect statement w.r.t. *Periplaneta americana*.

(1) The compound eyes are situated at the dorsal surface of the head
(2) The brain is represented by sub-oesophageal ganglion which supplies nerves to antennae and compound eyes
(3) Possess three ganglia in the thorax and six ganglia in the abdomen
(4) Exhibits mosaic vision with more sensitivity but less resolution

60) Given below are two statements :
Statement-I :- The head of the cockroach is connected with thorax by a short extension of the head known as the neck.
Statement-II :- When resting, the hind wings of the cockroach are covered by the forewings.
In the light of the above statements, choose the correct answer from the options given below :-

(1) Both statement-I and statement-II are false.
(2) Statement-I is true but statement-II is false.
(3) Statement-I is false but statement-II is true.
(4) Both statement-I and statement-II are true.

61) Which type of mouth parts present in cockroach -

(1) Biting and chewing

(2) Biting and Sucking

(3) Sucking and chewing

(4) Sucking and churning

62) Which of the following structure does not help in excretion in cockroach ?

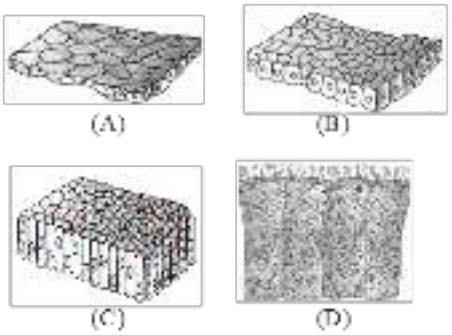
(1) Malpighian tubules

(2) Fat bodies

(3) Nephrocytes

(4) Phallic gland

63) See the following figure and identify the following simple epithelial tissues :-



	A	B	C	D
(1)	Squamous	Cuboidal	Columnar	Pseudo-stratified squamous (ciliated)
(2)	Pseudo stratified squamous	Cuboidal	Columnar	Ciliated columnar
(3)	Squamous	Cuboidal	Columnar	Ciliated columnar
(4)	Cuboidal	Squamous	Columnar	Ciliated columnar

(1) 1

(2) 2

(3) 3

(4) 4

64) How many of the following statements are related to bone ?
(A) It is specialised connective tissue having hard and pliable ground substance rich in Ca^{+2} salts
(B) It is the main tissue that provide structural frame to the body.
(C) It support and protects softer tissues and organs.
(D) The bone cells, osteocyte are present in the spaces called lacunae
(E) They also interact with smooth muscles attached to them to bring about movements.

(1) Two

(2) Three

(3) Four

(4) Five

65) **Assertion :-** Cells of adipose tissue are specialised to store carbohydrate.
Reason :- Excess of nutirent which is not used immediately converted in fat and is stored in the tissue.
Which of the following option is correct regarding given statements.

(1) Both (A) and (R) are true and (R) is correct explaination of (A)

(2) Both (A) and (R) true but (R) is not correct explaination of (A)

(3) (A) is true but (R) is false

(4) (A) is false but (R) is true

66) Match the column-I with column-II :

	Column-I		**Column-II**
(A)	Squamous epithelium	(1)	Present in bronchioles
(B)	Cuboidal epithelium	(2)	Present in lungs
(C)	Columnar epithelium	(3)	Present in stomach
(D)	Ciliated epithelium	(4)	Present in kidney

(1) A-2, B-4, C-3, D-1

(2) A-4, B-3, C-2, D-1

(3) A-3, B-2, C-1, D-4

(4) A-1, B-2, C-3, D-4

67) Which epithelium provides maximum protection against chemical and physical stress :-

(1) Epithelium of air sac

(2) Epithelium of proximal convoluted tubule

(3) Epithelium of buccal cavity

(4) Epithelium of fallopian tube

68) Which of the following is present in tip of nose and outer ear joint ?

(1) Cartilage

(2) Areolar tissue

(3) Smooth muscle

(4) Intercalated discs

69) Choose incorrect option with respect to type of tissue and their function :-

(1) Areolar - Present beneath the skin

(2) Tendon - Connect skin with muscle

(3) Ligament - Connect bone with bone

(4) Adipose - Specialised to store fat

70) From the list of given characters, which of them represent *Chondrichthyes* & *Osteichthyes* ?
(a) Placoid scales
(b) Operculum present
(c) Air bladder present
(d) Many of them are viviparous
(e) External fertilisation
(f) Notochord persistent throughout life.

	Chondrichthyes	**Osteichthyes**
(1)	a, b, d	c, e, f
(2)	a, d, f	b, c, e
(3)	b, c, d	a, e, f
(4)	b, c, e	a, d, f

(1) 1
(2) 2
(3) 3
(4) 4

71) Metameric segmentation is present in :

(1) Platyhelminthes, Annelida, Mollusca
(2) Annelida, Arthropoda, Mollusca
(3) Annelida, Arthropoda, Chordata
(4) Enchinodermata, Hemichordata, Chordata

72) Identify the correct match from the column-I, II and III.

	Column-I		**Column-II**		**Column-III**
1.	Adamsia	a.	Sea fur	i.	Medusa only
2.	Aurelia	b.	Sea anemone	ii.	Polyp and medusa both
3.	Obelia	c.	Sea gooseberry	iii.	no polyp and medusa
4.	Pleurobrachia	d.	Jelly fish	iv.	Polyp only

(1) 2-b-i 1-d-iv 3-a-ii 4-c-iii
(2) 1-b-i 2-d-iv 3-a-ii 4-c-iii
(3) 1-b-iv 2-d-i 3-c-iii 4-a-ii
(4) 2-d-i 1-b-iv 3-a-ii 4-c-iii

73) Choose the incorrect statements with respect to chordates -
(A) Presence of mid ventral, solid and double nerve cord.

(B) Presence of open circulatory system.
(C) Presence of ventral heart.
(D) Presence of paired pharyngeal gills slits.
(E) Triploblastic acoelomate animals.

(1) C and D only
(2) A, C and D only
(3) A, B and E only
(4) A, D and E only

74) Which of the following represent the correct combination without any exception ?

	Characteristic	Class
(1)	Porous body, choanocytes water, canal system	Porifera
(2)	Cnidocytes, tissue level organisation, gastrovascular cavity	Cnidaria
(3)	Flattened body exclusive sexual reproduction, internal fertilization	Platyhelminthyes
(4)	Exclusive sexual reproduction, exclusive marine comb plates	Ctenophora

(1) 1
(2) 2
(3) 3
(4) 4

75) **Assertion :-** Aves and mammals are homeothermal animals.
Reason :- Aves and mammals are able to maintain a constant body temperature.

(1) **Assertion** is correct but **Reason** is incorrect
(2) **Assertion** is incorrect but **Reason** is correct
(3) Both **Assertion** and **Reason** are correct and **Reason** is the correct explanation of **Assertion**
(4) Both **Assertion** and **Reason** are correct but **Reason** is NOT the correct explanation of **Assertion**

76) **Assertion (A) :-** Hemichordata are considered as a subphylum under phylum chordata.
Reason (R) :- Hemichordates have notochord in their collar region.

(1) A & R correct and R is correct explanation of A
(2) A & R both are incorrect
(3) A & R correct but R does not explain A
(4) A is correct but R is incorrect

77) **Statement-I :-** Echinoderms are triploblastic and coelomate animals.
Statement-II :- An excretory system is absent in Echinoderms.

(1) Both statement I & II are correct

(2) Both statement I & II are incorrect

(3) Only statement I is correct

(4) Only statement II is correct

78) **Statement-I :-** Hemidactylus is commonly known as Tree lizard.
Statement-II :- Hemidactylus is homoiothermous animal.

(1) Statement-I is correct and statement-II is incorrect

(2) Both statement-I and statement-II are correct

(3) Both statement-I and statement-II are incorrect

(4) Statement-I is incorrect and statement-II is correct

79) Match the column-I with column-II :

	Column-I		**Column-II**
(i)	Apis	(a)	Silkworm
(ii)	Bombyx	(b)	Honey bee
(iii)	Laccifer	(c)	King crab
(iv)	Limulus	(d)	Lac insect

(1) i-b, ii-a, iii-d, iv-c

(2) i-a, ii-b, iii-d, iv-c

(3) i-c, ii-b, iii-d, iv-a

(4) i-c, ii-a, iii-d, iv-b

80) Match the column-I (Classes) and Column-II (Example) :

	Column-I		**Column-II**
(a)	Chondrichthyes	(i)	*Myxine*
(b)	Osteichthyes	(ii)	*Pristis*
(c)	Amphibia	(iii)	*Salamandra*
(d)	Cyclostomata	(iv)	*Clarias*

	(a)	(b)	(c)	(d)
(1)	ii	iv	iii	i
(2)	iv	ii	iii	i
(3)	i	ii	iii	iv
(4)	iv	i	iii	ii

(1) 1

(2) 2

(3) 3

(4) 4

81) Which one of the following option is correct for these four animals
Fasciola, Earthworm, Psittacula and Camelus ?

(1) Camelus is poikilothermic while the remaining are homeothermic.

(2) All are free living organisms.

(3) All possess closed circulatory system.

(4) All are bilaterally symmetry.

82) Which of the following is incorrect about birds?

(1) Air sacs are connected to lungs which help in respiration.

(2) Hind limb possess scales and are modified for walking, swimming or clasping

(3) Separate sexes, internal fertilization, oviparous and direct development

(4) Endoskeleton consists of feathers, scales, beak and claws.



83) Identify the Animal and find out the incorrect option for given animal :

(1) Presence of bioluminescence.

(2) Presence of ten comb plates

(3) Reproduction only sexual

(4) Tissue level organisation

84) Select the incorrect statements with respect to cyclostomes :
(a) They are ectoparasites on some fishes.
(b) Migrate for spawning to marine water, after spawning within few days, they die.
(c) Cranium is cartilaginous but vertebral column is bony.
(d) Their body is devoid of scales but unpaired fins are present

(1) b, c and d

(2) a and d

(3) b and c

(4) c and d

85) Which of the following is dioecious animal ?

(1) Neries

(2) Earthworm

(3) Leech

(4) Fasciola

86) In which of the following animals, Alimentary canal is complete with a well develop muscular pharynx ?

(1) Ascaris

(2) Anopheles

(3) Apis

(4) Astarias

87) In which phylum feather like gills are present for excretion :-

(1) Mollusca

(2) Arthopoda

(3) Porifera

(4) Ctenophora

88) Notochord extends from head to tail region and persists throughout their life in :-

(1) Ascidia

(2) Salpa

(3) Doliolum

(4) Amphioxus

89) Which of the following animals are also called tunicates ?

(1) Urochordata

(2) Cephalochordata

(3) Cyclostomata

(4) Vertebrata

90) In which of the following animals possess high regeneration capacity ?

(1) Taenia

(2) Fasciola

(3) Planaria

(4) Schistosoma

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	4	4	3	4	2	2	2	2	4	4	4	4	4	3	3	4	4	4	1	3
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	3	2	3	2	3	2	2	2	2	3	1	3	2	2	1	2	3	4	2	3
Q.	41	42	43	44	45															
A.	1	3	1	1	3															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	4	1	2	3	1	1	3	1	1	3	2	1	1	4	4	4	2	2	1	1
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	1	4	1	1	3	4	1	2	1	3	3	4	1	3	4	2	3	1	3	4
Q.	86	87	88	89	90															
A.	2	3	1	4	1															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	1	2	1	1	3	1	2	4	3	3	1	4	4	1	4	3	2	4	4	4
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	2	2	2	4	1	3	2	3	2	1	1	1	1	2	4	4	2	4	2	2
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	3	3	3	1	3	2	1	1	2	1	3	2	2	4	1	2	3	1	2	3
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	1	4	3	2	4	1	3	1	2	2	3	4	3	3	3	2	1	3	1	1
Q.	171	172	173	174	175	176	177	178	179	180										
A.	4	4	2	3	1	1	1	4	1	3										

SOLUTIONS

PHYSICS

1)

D = (4.23 ± 0.01)cm
d = (3.87 ± 0.01) cm

$t = \frac{D}{2} - \frac{d}{2}$

$= \frac{4.23}{2} - \frac{3.87}{2}$

t = 0.18

$\Delta t = \frac{\Delta D}{2} + \frac{\Delta d}{2}$

$= \frac{0.01}{2} + \frac{0.01}{2}$

Δt = 0.01

 t = (0.18 ± 0.01)cm

2) $\frac{\Delta d}{d} \times 100 = \left(\frac{0.003}{0.3} \times 100\right)$

$+ \left(2 \times \frac{0.005}{0.5} \times 100\right) + \left(\frac{0.06}{6} \times 100\right)$

= 4%

3) $\frac{10,569 - 958.82 - 1.8 \times 10^{-1}}{0.9950 + 8.60 + 0.015} = 1000$

$= 1.00 \times 10^3$

1.0569	0.9950
− 958.82	+ 8.60
− 0.18	+ 0.015
9610.00	9.6100

4 significant figure 3 significant figure

So in Answer significant figure must be 3

4) Ans. (4)

$[F] = \left[\frac{Cr}{V^2}\right]$

$\left[M^1L^1T^{-2}\right] = \left[\frac{CL}{L^2}T^2\right]$

$[C] = [M^1L^2T^{-4}]$

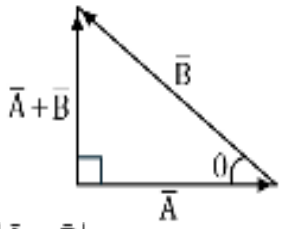
5) $[A] = \frac{[v]}{T^3} = LT^{-4}$, $[B] = \frac{[v]}{T}$

$a = 0 \Rightarrow \frac{dv}{dt} = 0 \Rightarrow 3At^2 - B = 0$

6)

In (B) volume = L^3
length = L
as volume is completely derival from length, it is a derived quantity

7)



$|\bar{A} + \bar{B}| = x$
$|\bar{B}| = 2x$
$\text{Sin}\,\theta = \frac{x}{2x} = \frac{1}{2}$
$\theta = 30^\circ$
angle b/w $\bar{A}$ and $\bar{B}$ = 180° – 30° = 150°

8)

On yz plane x component will become zero.

$\vec{P}_{yz} = 8\hat{j} + 6\hat{k}$
$|\vec{P}_{yz}| = \sqrt{6^2 + 8^2} = 10$

9) L.C = 1mm

$\text{L.C} = \frac{\text{Pitch}}{\text{Number of division on the circular scale}}$

$8 \times 10^{-6} = \frac{10^{-3}}{N}$

$N = \frac{10^{-3}}{8 \times 10^{-6}} = \frac{10^3}{8} = 125$

10) $F_A = \frac{\mu mg}{\cos 30^\circ - \mu \sin 30^\circ}$

$F_B = \frac{\mu mg}{\cos 60^\circ + \mu \sin 60^\circ}$

$\frac{F_A}{F_B} = \frac{\cos 60^\circ + \mu \sin 60^\circ}{\cos 30^\circ - \mu \sin 30^\circ}$

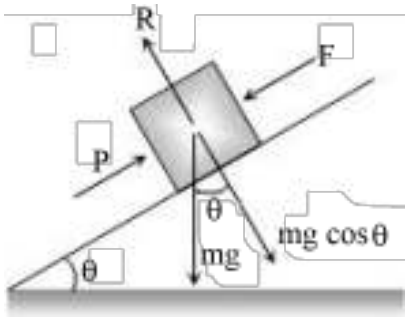
$$\frac{F_A}{F_B} = \frac{2}{\sqrt{3}}$$

11)

Net downward force = Weight – Friction
 ma = 25 × 9.8 – 2
$\Rightarrow a = \frac{25 \times 9.8 - 2}{25} = 9.72 \text{ m/s}^2$

12)

Net force along the plane = P – mg sin θ = 750 – 500 = 250 N
Limiting friction = $F_1 = \mu_s R = \mu_s$ mg cos θ
= 0.4 × 102 × 9.8 × cos 30 = 346 N
As net external force is less than limiting friction therefore friction on the body will be 250 N.



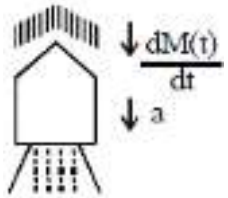
13) $\frac{dm(t)}{dt} = bv^2$

$F_{thrust} = v\frac{dm}{dt}$

Force on satellite = $-\vec{v}\frac{dm(t)}{dt}$

$M(t)a = -v(bv^2)$

$a = -\frac{bv^3}{M(t)}$



14)

(3) Force on the football $F = m\frac{dv}{dt}$

$F = \frac{m(v_2 - v_1)}{dt} = \frac{0.5 \times (10 - 0)}{1/50} = 250 \text{ N}$

15) $\frac{1}{v}-0=\frac{\frac{1}{2}-0}{4-0}(x-0)$

$\frac{1}{v_{12}}=\frac{x}{8}\Rightarrow v=\frac{8}{x}\Rightarrow \frac{dx}{dt}=\frac{8}{x}$

$\int_4^{12} xdx = 8\int_0^t dt$

$\left[\frac{x^2}{2}\right]_4^{12} = 8t \Rightarrow t = 8 \text{ sec}$

16)

Initial vertical velocity and vertical acceleration of particle is same for both w.r.t car and w.r.t ground. Thus time of flight remains same.

17) Distance = $\int_0^1 |v|\,dt$ = 1 unit

18) $R=\frac{V^2(2\sin\theta\cos\theta)}{g}$

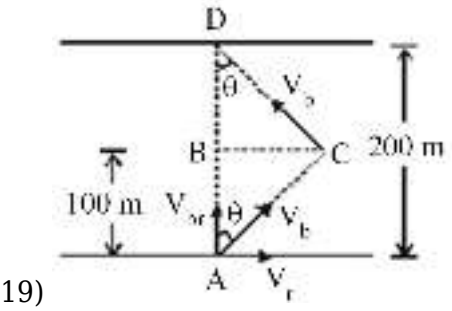
$t=\frac{V\sin\theta}{g}\Rightarrow V=\frac{gt}{\sin\theta}$

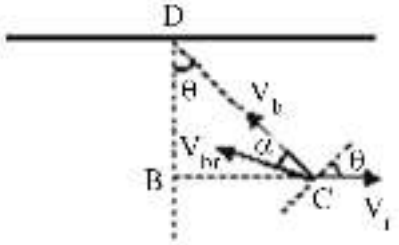
$\Rightarrow \quad R=\frac{g^2t^2}{\sin^2\theta}.\frac{2\sin\theta\cos\theta}{g}$

$\tan\theta=\frac{2gt^2}{R}=\frac{20t^2}{R}$

$\cot\theta=\frac{R}{20t^2}$

19)





$V_{br} = 4$ m/s, $V_r = 2$ m/s

$$\tan\theta = \frac{BC}{AB} = \frac{|V_r|}{V_{br}} = \frac{2}{4} = \frac{1}{2}$$

In this case, V_b should be along CD.

$$V_r\cos\theta = V_{br}\sin\alpha$$

$$2\left(\frac{2}{\sqrt{5}}\right) = 4\sin\alpha$$

or $$\sin\alpha = \frac{1}{\sqrt{5}}$$

$$\alpha = \theta = \tan^{-1}\left(\frac{1}{2}\right)$$

Let, time taken if he would have denied be t_1.

$$t_1 = \frac{200}{|V_{br}|} = \frac{200}{4} = 50\text{ s}$$

Let, time taken when not denied be t_2.

$$DC = BD\sec\theta = (100)\frac{\sqrt{5}}{2} = 50\sqrt{5}\text{m}$$

$$|V_b| = |V_{br}|\cos\alpha - |V_r|\sin\theta$$

$$= 4\left(\frac{2}{\sqrt{5}}\right) - 2\left(\frac{1}{\sqrt{5}}\right) = \frac{6}{\sqrt{5}}\text{m/s}$$

$$t_2 = \frac{t_1}{2} + \frac{DC}{|V_b|} = 25 + \frac{50\sqrt{5}}{\frac{6}{\sqrt{5}}} = \frac{200}{3}\text{s}$$

$$\frac{t_2}{t_1} = \frac{4}{3}$$

20) $$\frac{da^2}{dv} = \tan 37^\circ$$

$$2a\frac{da}{dv} = \frac{3}{4}$$

$$\frac{dv}{dt} \times \frac{da}{dv} = \frac{3}{8}$$

21) $\vec{v}_{12} = \vec{v}_1 - \vec{v}_2$

$|\vec{v}_{12}|_{max} = |\vec{v}_1| + |\vec{v}_2|$

$|\vec{v}_{12}|_{min} = |\vec{v}_1| - |\vec{v}_2|$

22) $\vec{v}_W = \frac{v}{\sqrt{2}}\hat{i} + \frac{v}{\sqrt{2}}\hat{j}$, $\vec{v}_m = 0\,\hat{i} + at\,\hat{j}$

$\vec{v}_{WM} = \frac{v}{\sqrt{2}}\hat{i} + \left(\frac{v}{\sqrt{2}} - at\right)\hat{j} \Rightarrow \frac{v}{\sqrt{2}} - at = 0$ get

$t = 2$

23) $2t = \frac{2u}{g}$, $h = u \times \frac{t}{2} - \frac{1}{2}g\left(\frac{t}{2}\right)^2$

24) $\frac{at\hat{i}+bt^2\hat{j}}{t}=a\hat{i}+bt\hat{j}$

25) $\alpha-\beta t^2=0,\ t=\sqrt{\frac{\alpha}{\beta}}$

$x=\alpha t-\frac{\beta t^2}{3}=t\left(\alpha-\frac{\beta t^2}{3}\right)=\frac{2}{3}\alpha\sqrt{\frac{\alpha}{\beta}}$

26) $\vec{u_A}=10\hat{i},\ \vec{a_A}=-g\hat{j}$

$\vec{u_B}=25\cos 37^o\hat{i}+25\sin 37^o\hat{j}=20\hat{i}+15\hat{j}$

$=\vec{a_B}=-g\hat{j}$

$\vec{a_{AB}}=\vec{0}$

$\vec{V_{BA}}=10\hat{i}+15\hat{j}=$ constant

$\left|\vec{V_{BA}}\right|=\sqrt{325}$

$d=vt$

$=\sqrt{325}\times 2$

$=10\sqrt{13}$ m

27) $t^2=x^2-1$

$x^2=1+t^2$

$x\frac{dx}{dt}=t$

$\Rightarrow\left(\frac{dx}{dt}\right)^2+x\frac{d^2x}{dt^2}=1$

$\Rightarrow\frac{xd^2x}{dt^2}=1-\left(\frac{dx}{dt}\right)^2=1-\left(\frac{t}{x}\right)^2=\frac{x^2-t^2}{x^2}=\frac{1}{x^2}$

$\Rightarrow a=\frac{d^2x}{dt^2}=\frac{1}{x^3}$

28) Maximum height $H=\frac{u^2\sin^2\theta}{2g}$

$\Rightarrow h_1=\frac{(20)^2\sin^2\theta}{2g}=20\sin^2\theta$ &

$h_2=\frac{(20)^2\sin^2(\pi/2-\theta)}{2g}=20\cos^2\theta$

$h_2-h_1=20\ [\cos^2\theta-\sin^2\theta]\ =10$

$\Rightarrow 20\cos 2\theta=10$

$\Rightarrow\cos 2\theta=\frac{1}{2}\Rightarrow 2\theta=60°$

$\Rightarrow\theta=30°$ and $\theta=90°-\theta=60°$

29) Consecutive divisions of main scale are

5.10 cm & 5.15 cm
⇒ 1 MSD = 5.15 – 5.10 = 0.05 cm
50 divisions of vernier scale = 2.45 cm
1 Division = 1 VSD
$= \frac{2.45}{50}\text{cm}$
= 0.049 cm
Least count of vernier calliper = 1 MSD – 1 VSD
= (0.05 – 0.049)cm
= 0.001 cm
 Diameter = MSR + VSR – ZE
= 5.10 cm + 24 (0.001 cm) – 0
= 5.124 cm

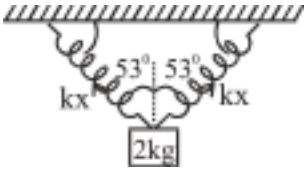
30) Density of material = $4\,\frac{g}{cm^3} = \frac{4\times 1000}{100}\frac{100g}{(10cm)^3} = 40\,\frac{(100g)}{(10cm)^3}$

31)

L.C. = $\frac{\text{Pitch}}{\text{No. of division}} = \frac{1\,mm}{50} = 0.02$ mm
Zero error = – (50 – 44) × 0.02 = – 0.12 mm
Reading = MSR + VSR × LC – 2E
= 3 + 26 × 0.02 – 2E
= 3 + 0.52 + 0.12 = 3.64 mm

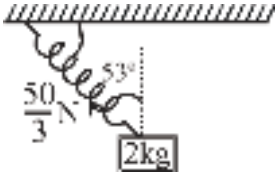
32) $\vec{a} = \frac{d\vec{v}}{dt} = 5\hat{i} + 2\hat{j}$
$\vec{F} = m\vec{a} = ma_x\hat{i} + m(g + a_y)\hat{j}$
$\left|\vec{F}\right| = m\sqrt{a_x^2 + (g + a_y)^2} = 26\,N$

33)



2 kx cos 53º = 20
$2\,kx \times \frac{3}{5} = 20$
$kx = \frac{50}{3}N.$
x direction

$a_x = \frac{\frac{50}{3}\sin 53}{2} = \frac{20}{3}m/s^2$



$$a_y = \frac{\frac{50}{3}\cos 53 - 20}{2} = 5\ m/s^2$$

$$a = \sqrt{\left(\frac{20}{3}\right)^2 + 5^2} = \frac{25}{3} m/s^2$$

34) $F\sin 37° - mg - \mu F\cos 37° = ma$

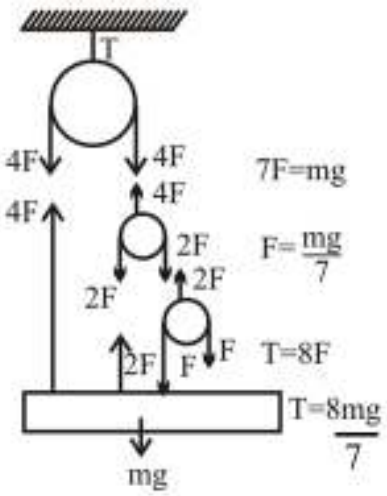
$\Rightarrow \quad a = 20\ m/s^2$

$$75 \times \frac{3}{5} - 1 \times 10 - 0.25 \times 75 \times \frac{4}{5} = 1 \times a$$

$45 - 10 - 15 = 1 \times a$

$20 = a$

35)



36) The maximum tension = $30 \times 10N = 300N$

$T - mg = ma. \quad 300 - 10 \times 10 = 10a$

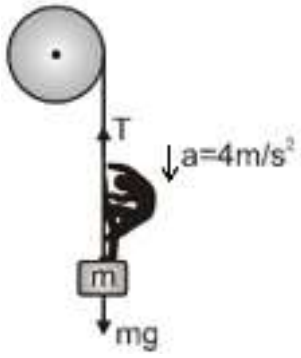
$\Rightarrow a = 20\ m/s^2$ [Maximum value]

when the mass is raised, $u = 0$, $a = 20\ m/s^2$.

$S = 10m, \ t = ?$

$$10 = \frac{1}{2} \times 20 .t^2 \ \Rightarrow \ t = 1s.$$

37)



$mg - T = ma$

$60 \times 10 - 360 = 60\ a$

i.e. slip through the rope with a minimum acceleration $4m/s^2$.

38) $\vec{F} = -\frac{\partial U}{\partial x}\hat{i} - \frac{\partial U}{\partial y}\hat{j} = 7\hat{i} - 24\hat{j}$

$\therefore a_x = \frac{F_x}{m} = \frac{7}{5} = 1.4\ ms^{-2}$

and $\therefore a_y = \frac{F_y}{m} = -\frac{24}{5} = -4.8\ ms^{-2}$

$\therefore v_x = a_x t = 1.4 \times 2 = 2.8\ ms^{-1}$

and $v_y = a_y t = 4.8 \times 2 = 9.6\ ms^{-1}$

$\therefore v = \sqrt{v_x^2 + v_y^2} = 10\ ms^{-1}$

39) $a = \left(\frac{2m - m}{2m + m}\right) g = \frac{g}{3}$

$T = \frac{2(m)(2m)}{m + 2m} g = 16$

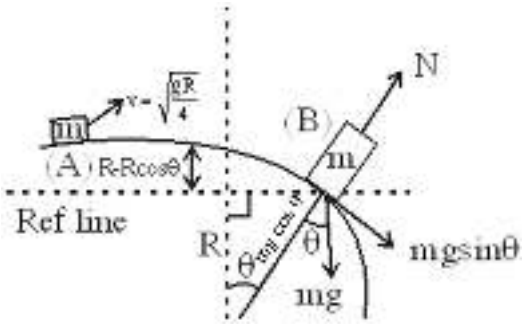
$\Rightarrow m = \frac{12}{g}$

$v = u + at = 0 + \left(\frac{g}{3}\right)(1)$

$-\Delta U = +\Delta KE = \frac{1}{2}(m + 2m)v^2$

$-\Delta U = \frac{1}{2}(3m)\frac{g^2}{9} = \frac{3}{2} \times \frac{12}{g} \times \frac{g^2}{9}$

$= 2g = 19.6\ J$



40)

If the Block of mass (m), looses contact at point-B, then N = 0.

Apply MEC, A → B

$PE_i + KE_i = KE_f + PE_f$

$Mg\ (R - R\cos\theta) + \frac{1}{2}m\left(\sqrt{\frac{gR}{4}}\right)^2 = \frac{1}{2}mv_B^2 + 0$

$gR - gR\cos\theta + \frac{1}{2}\frac{gR}{(4)} = \frac{V_B^2}{2}$

$\therefore V_B = \sqrt{\frac{8gR - 8gR\cos\theta + gR}{4}}$

Equation at point B

$mg\cos\theta - N = \frac{mV_B^2}{R}$

$mg\cos\theta = m\left[\frac{9gR - 8gR\cos\theta}{4R}\right]$ (N = 0)

$12\cos\theta = 9$

$\cos\theta = \frac{9}{12} = \frac{3}{4}$

$\theta = \cos^{-1}\left(\frac{3}{4}\right)$

41) $F = 8t$

$\Rightarrow m\frac{dv}{dt} = 8t \Rightarrow m\int_0^v dv = 8\int_0^t t\,dt$

$\Rightarrow mv = 4t^2 \Rightarrow m\frac{dx}{dt} = 4t^2$

$\Rightarrow dx = \frac{4t^2}{m}dt = 2t^2dt$

$\Rightarrow W = \int F dx = \int_0^2 (8t \times 2t^2)dt = 64J$

or

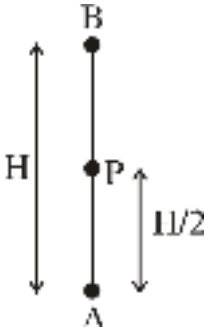
$\Rightarrow F = 8t \Rightarrow m\frac{dv}{dt} = 8t \Rightarrow dv = \frac{8t\,dt}{m}$

$\Rightarrow \int_0^v dv = \int_0^t 4t\,dt \Rightarrow v = 2t^2$

So, $V_{t=0} = 0$ & $V_{t=2s} = 8$ m/s

$W = \frac{1}{2}m\left(v^2 - O^2\right) = \frac{1}{2} \times 2 \times 8^2 = 64$ J

42) **Case-I :**



When projected with speed v by COME law

$\frac{1}{2}mv^2 = mgh$ ....(1)

Case-II :

When projected with speed 2v

$v^2 = u^2 + 2as$

$v_P^2 = (2v)^2 - 2g\frac{H}{2}$

$v_P^2 = 4v^2 - gH$ ....(2)

So that KE at point P

$\frac{1}{2}mv_P^2 = \frac{1}{2}m[4v^2 - gH]$

$\frac{1}{2}mv_P^2 = 4\left[\frac{1}{2}mv^2\right] - \frac{mgH}{2}$ ...(3)

By equation (1) & (3)

$$(KE)_P = \frac{1}{2}mv_P^2 = \frac{7}{2}mgH$$

$$(PE)_P = \frac{mgH}{2}$$

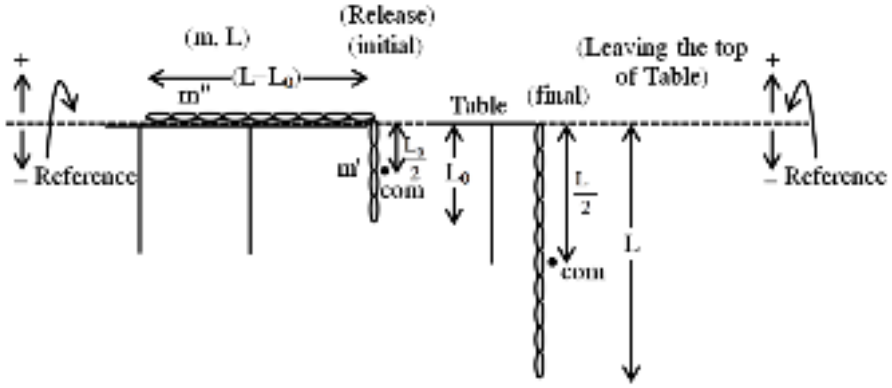
So that

$$\frac{(PE)_P}{(KE)_P} = \frac{1}{7}$$

43) $P = \frac{dw}{dt} = \frac{dk}{dt}$

$$\int dk = \int_4 Pdt$$

$$\Delta KE = \int_2 (3t^2 - 2t + 1)dt$$



44)

$$m' = \frac{m}{L}L_0$$

Applying M.E.L.

$KE_i + PE_i = KE_f + PE_f$

$$0 + m''g \times 0 - m'g\frac{L_0}{2} = \frac{1}{2}mv^2 - mg\frac{L}{2}$$

$$-\left(\frac{m}{L}L\right)g\frac{L_0}{2} = \frac{1}{2}mv^2 - mg\frac{L}{2}$$

$$-\frac{mgL_0^2}{2L} + \frac{mgL}{2} = \frac{mv^2}{2}$$

$$-\frac{gL_0^2}{L} + gL = v^2$$

$$\sqrt{-\frac{gL_0^2 + gL^2}{L}} = v$$

$$\therefore v = \sqrt{\frac{g}{L}(L^2 - L_0^2)}$$

45)

$$S = vt = \sqrt{2g(H-h)}\sqrt{\frac{2h}{g}} = 2\sqrt{h(H-h)}$$

For S-max, h(H-h) → max

$$\frac{d(Hh-h^2)}{dh} = 0$$

$H-2h = 0 \Rightarrow h = \frac{H}{2}$

$S = \sqrt{4\frac{H}{2}\left(H-\frac{H}{2}\right)}$

$S = H$

CHEMISTRY

46) $Fe(OH)_2(s) \rightleftharpoons Fe^{2+}(aq.) + 2OH^-(aq.)$

$t_{eq.}$ s 10^{-1} M

$K_{sp} = [Fe^{2+}][OH^-]^2$

$8 \times 10^{-16} = (s)^1(10^{-1})^2$

47) According to bronsted lowry concept, proton donor act as acid.

48) $NH_4^+(aq.) \rightleftharpoons NH_3(aq.) + H^+(aq.) : K_1$

$HCN(aq.) \rightleftharpoons H^+(aq.) + CN^-(aq.) : K_2$

$NH_4^+(aq.) + CN^-(aq.) \rightleftharpoons NH_3(aq.) + HCN(aq.)$

$K_{eq} = \frac{K_1}{K_2} = \frac{5.76}{4.8} = 1.2$

49)

NH_4OH should be greater then HCl

$NH_4OH + HCl \rightarrow NH_4Cl + H_2O$

0.1 0.05 0 0

0.05 0 0.05

It is a basic buffer

50)

mili mole of $OH^- = 30 \times 0.1 \times 2$

$= 6$

mili mole of $H^+ = 20 \times 0.05 = 1$

$[OH^-] = \frac{6-1}{20+30} = \frac{1}{10} = 0.1\ M$

51) $[OH^-]$ in solution $= 10^{-7} \times 100 = 10^{-5}$

pOH = 5, pH = 9

52) Multiplying first reaction by 2 and then subtracting the second reaction gives the required reaction,

$\Delta H = -230 \times 2 + 300$
$= -160$ kJ

53)

$\Delta H_{rxn} = \Sigma B.E._{(reactants)} - \Sigma B.E._{(products)}$
$= 4 \times 414 + 2 \times 243 - [2 \times 414 + 2 \times 150 + 2 \times 432]$
$= 150$ kJ

54) $-12250x - 13000(1 - x) = -12500$
$750x = 500 \Rightarrow x = 2/3$ and $y = 1/3$
So, $\frac{x}{y} = \frac{2}{1}$

55) $\Delta H^{o}_{comb}(H_2) = \Delta H^{o}_{f}(H_2O)$
Calorific value $= \frac{\Delta H^{o}_{comb}(H_2)}{Mw_{H_2}}$
$-143 = \frac{\Delta H^{o}_{f}(H_2O)}{2}$
$\Delta H^{o}_{f}(H_2O) = -286$ kJ mol^{-1}

56)

For a spontaneous reaction entropy of universe increases and reached to the maximum value when equilibrium is reached.

57)

$\Delta U = q + W$
Here $q = 0$
$W = -P(V_2 - V_1)$
$= -1(10 - 2) = -8$
$\Delta U = -8$ L atm

58)

Max. Work = reversible work
$W = -2 \times 8.314 \times 300 \times 2.303 \log \frac{100}{10}$
$= -11488$ J
$= -11.5$ kJ

59)

Theory based

60)

Conc. of A and B decreases and conc. of C increases and conc. of B decreases more faster than A.

61)

1 gm-ion = 1 mole Al^{+3} ion charge = $1 \times 3 \times N_A \times e$ coulomb

62)

Mass of 20 molecule of $SO_3 = 20 \times 80$
$= 1600$
Number of O_2 molecule in 1600 weight
$$= \frac{1600}{32} = 50$$

63) $$C = \frac{40.7}{12} = 3.3916$$
$$H = \frac{5.1}{1} = 5.1$$
$$O = \frac{54.2}{16} = 3.387$$
Their simple ratio comes to be 2 : 3 : 2.
So the empirical formula = $C_2H_3O_2$.

64)

mole of $MnO_2 = \frac{5}{87}$
mole of HCl required = $\frac{5}{87} \times 4$
mass of HCl = $\frac{5 \times 4}{87} \times 36.5 = 8.39g$

65) (I) 0.5 mole O_3 = 24 g O_3;
(II) 0.5 g atom of oxygen = 8 g
(III) $\frac{3.011 \times 10^{23}}{6.022 \times 10^{23}} \times 32 = 16$ g O_2
(IV) $\frac{5.6}{22.4} \times 44gCO_2 = 11$ gCO_2

66)

L.R.
2A + 3B → 5C
5 10

$\frac{5 \times 5}{2} = \frac{25}{2}$

0 2.5

$4C + 7D \rightarrow 9E + 6F$

$\frac{25}{2}$ ↓

$\frac{25}{2} \times \frac{6}{4} = \frac{75}{4}$

$3F + 4G \rightarrow 8H$

$\frac{75}{4}$ $\frac{75}{4} \times \frac{8}{3} = 50$

67)

Mass of metal (in g) which combined with 8 g of oxygen in the formation of metal oxide, is called it's equivalent weight.

68) We know that $K_P = K_C(RT)^{\Delta ng}$ and $\Delta n = 2 - 1 = 1$

$\therefore K_P = K_C \times RT = 0.4 \times 0.082 \times 350 = 11.48$ atm.

69)

For reaction $\Delta n_g > 0$ when volume increases reaction proceed in forward direction.

70) $S(s) + O_2(g) \rightleftharpoons SO_2(g)$ $K_1 = 10^{52}$ ...(1)

$2S(s) + 3O_2(g) \rightleftharpoons 2SO_3(g)$ $K_2 = 10^{129}$ ...(2)

$2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$ $K_3 = x$

multiplying equation (1) by 2;

$2S(s) + 2O_2(g) \rightleftharpoons 2SO_2(g)$ $K_1' = 10^{104}$ ...(3)

⇒ Substracting (3) from (2); we get

$2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$;

$K_{eq} = 10^{(129 - 104)} = 10^{25}$

71) $Q = \frac{[SO_2][CO_2]}{[SO_3][CO]} = \frac{2 \times 2}{2 \times 2} = 1$

K < Q reaction proceed in backward direction

$SO_3 + CO \rightleftharpoons SO_2 + CO_2$

2+x 2+x 2–x 2–x

$n(SO_3) + n(CO_2) = 2 + x + 2 - x = 4$

$\frac{n(SO_2)}{n(CO)} = \frac{2-x}{2+x} < 1$

72) $x \rightleftharpoons 2y$

1-α 2α

$K_P = \frac{\left(\frac{2\alpha}{1+\alpha}\right)^2 \times P}{\left(\frac{1-\alpha}{1+\alpha}\right)}$

$A \rightleftharpoons B + C$

$1-\alpha \quad \alpha \quad \alpha$

$$K_P' = \frac{\left(\frac{\alpha}{1+\alpha}\right)^2 P'}{\left(\frac{1-\alpha}{1+\alpha}\right)}$$

$K_P = 2K_P'$

$$4P = 2P^1 \Rightarrow \frac{P}{P'} = \frac{1}{2}$$

73) $K_f[\text{React}]^{\text{S.C.}} = 5\,K_b[\text{Products}]^{\text{S.C.}}$

$$\frac{[\text{Products}]^{\text{S.C.}}}{[\text{Reactant}]^{\text{S.C.}}} = \frac{K_f}{5\,K_b}$$

$$Q = \frac{1}{5}K_{eq}$$

$$Q = \frac{25}{5} = 5$$

74)

	A	+	3B	⇌	2C
initially	$\frac{1}{10}$		$\frac{1}{10}$		0
at eq.	0.1 – x		0.1 – 3x		2x

[A] > [B]

75) $\Delta H_{sol} = \Delta H_{Lattice} + \Delta H_{hyd.}(Na^+) + \Delta H_{hyd.}(Cl^-)$

$= 788 - 300 - 484 = 4$

76) Theory based.

77) Balanced chemical reaction is

$2MnO_4^- + 3CN^- + H_2O \rightarrow 2MnO_2 + 3CNO^- + 2OH^-$

78) $NaOH + NaH_2PO_3 \rightarrow Na_2HPO_3 + H_2O$

$2 \qquad 20 \times 0.1$

$NaOH + NaHCO_3 \rightarrow Na_2CO_3 + H_2O$

$6 \qquad 60 \times 0.1$

79)

mili Eq. of $KMnO_4$ = M × V(mL) × V.F.

= 0.1 × 20 × 5 = 10

mili Eq. of $H_2C_2O_4$ = 0.2 × 25 × 2 = 10

80) $N^{3-} \rightarrow N^{2+} + 5e \qquad \Rightarrow E = M/5$

$6e^- + 2N^{3-} \rightarrow (\overset{0}{N_2})_2$ ⇒ E = M/3

81) $\overset{(+3)(-1)}{CrI_3} \rightarrow \overset{+6}{Cr_2}O_7^{-2} + \overset{(+7)}{IO_4^-}$

n-factor ⇒ Cr = (+ 3 to + 6) = $3e^-$ loss

I = (– 1 to + 7) × 3 = 24 e^- loss

n-factor = 27

⇒ $E = \frac{M}{27}$

82) n-factor for $FeSO_4$ ⇒ Fe (+2 to + 3) × 1 = 1

n -factor for FeC_2O_4 ⇒

Fe(+2 to + 3) × 1 = $1e^-$ loss

C(+3 to + 4) × 2 = $2e^-$ loss = 3

Equivalent of FeC_2O_4 = Equivalent of $KMnO_4$

⇒ x mol × 1 = y mol × 5 .... (i)

Similarly,

Equivalent of FeC_2O_4 = Equivalent of $KMnO_4$

⇒ x mol × 3 = (mol of $KMnO_4$ = ?) × 5 ...(ii)

⇒ Mol of $KMnO_4$ required = 3y

83) At high altitude, external pressure decreases

84) Rate of evaporation is greater than rate of condensation, when equilibrium is reached both rate becomes equal.

85) * Conversion of water into steam above 373 K is spontaneous process at 1 atm

* $\Delta G = \Delta H^\circ$ – TΔS = –TΔS = +ve

* Ice and water maintain equilibrium at 1 atm and 273 K

* $\Delta S_{rev,\ adia} = 0$ ⇒ ΔG = ΔH

86) * BOH + HCl → BCl + H_2O

10 m mol 2.5 m mol

7.5 0 2.5

$pOH = pk_b + \log\frac{1}{3}$ ⇒ $pH = pk_w - pk_b - \log\frac{1}{3}$

* BOH + HCl → BCl + H_2O

10 m mol 5 m mol

5 0 5

$pOH = pk_b$ ⇒ $pH = pk_w - pk_b$

* BOH + HCl → BCl + H_2O

10 m mol 7.5 m mol

2.5 0 7.5

$pOH = pk_b + \log 3$ ⇒ $pH = pk_w - pk_b - \log 3$

* BOH + HCl → BCl + H_2O

10 m mol 10 m mol

0 0 10

$[BCl] = \frac{10}{200} = \frac{1}{20}$

$pH = \frac{1}{2}(pk_w - pk_b - \log C)$

87) $NH_4Cl \rightarrow NH_4^+ + Cl^-$

$NH_4^+ + H_2O \rightleftharpoons NH_4OH + H^+$

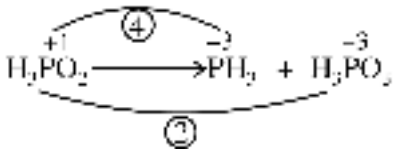
$[Cl^-] = [NH_4Cl] = 10^{-2}$ M

$[NH_4^+] < 10^{-2}$ due to hydrolysis

it is salt of SAWB so solution is acidic

88) For phenolphthalein pH colour change range is 8.2 to 10.2

89) On dilution solubility remains constant but moles increases.

90) 

$v.f = \frac{4 \times 2}{4+2} = \frac{4}{3}$

Eq. wt. $= \frac{M_w}{v.f.}$

BIOLOGY

91) NCERT XI Pg. # 4

92) NCERT-XI, Pg. # 8

93) NCERT-XI, Pg. # 29

94) NCERT-XI, Pg. # 17

95) NCERT XI Pg. # 19

96) NCERT-XI Pg.#20

97) NCERT-XI, Pg. # 11

98) NCERT-XI Pg#24

99) NCERT-XI, Pg. # 91

100) NCERT-XI, Pg. # 26

101) NCERT, Pg. # 26

102) NCERT Pg. # 26

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106)

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107) NCERT, Pg. # 17

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113) NCERT-XI, Pg. No. # 16-17

114)

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115) XI NCERT Pg. No. # 35, 36

116) NCERT Pg. # 24

117) NCERT, Pg. # 25, 28, 29, 32, 33

118) NCERT, Pg. # 24

119) NCERT, Pg. # 30

120)

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121)

XI NCERT Pg. No. # 10,11

122)

NCERT-XI, Pg. # 8

123) NCERT-XI Pg.#8

124) NCERT-XI Pg.#11
Hint: C, D & E correct.

125)

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126) XI NCERT Pg. No. # 17, 18
Mosses, ferns, gymnosperms, angiosperms.

127)

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128) Module

129) NCERT Pg. # 14

130) NCERT Pg. # 13, 14

131) NCERT-XI, Pg. 38-39

132)

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133)

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134)

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135)

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136)

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137) NCERT-XI, Pg. # 83

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139)

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140) NCERT (XIth) Pg. # 79

141)

Question Explanation: The question asks to match frog structures/functions (Column I) with their correct descriptions (Column II).

Concept: The question is based on frog .

Solution: The correct answer is Option 3 – A-iii, B-i, C-ii, D-iv

A. (A) Cutaneous respiration → (iii) Aquatic respiration (skin helps in respiration, especially in

water).

B. (B) Tympanum → (i) Receive sound signals (acts as the eardrum).
C. (C) Copulatory pad → (ii) 1st digit of fore-limbs (helps male frog hold female during amplexus).
D. (D) Ureotelic → (iv) Excrete urea (frogs excrete nitrogenous waste mainly as urea).

Final Answer : Option 3

142)

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143)

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144)

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145)

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146) Module - 4 Page No. # 123

147)

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148)

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149)

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150)

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151)

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152) Module - 4, Page No. # 130

153) Module - 4, Page No. # 82, 83

154)

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155) Module - 4

156) Module - 4, Page No. # 81, 82

157)

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158) Module - 4, Page No. # 94

159) Module - 4, Page No. # 90

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161)

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162)

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163)

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164)

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165) NCERT, Pg. # 49, 4.2.11.6

166) NCERT, Pg # 45

167) NCERT, Pg # 45

168) NCERT Page No. # 49

169) NCERT Page No. # 34

170) NCERT Page No. # 46, 47

171)

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172)

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173) NCERT XI Page No. # 40

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175)

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178) NCERT XI Page No. # 42

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